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Thermal management of thermoelectric generators: A thermal-resistance-chain perspective☆

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ABSTRACT

Thermoelectric generators (TEGs) provide a compact route for converting waste and low-grade heat into electricity, but their practical output remains far below the level expected from material properties alone. The key limitation is that the nominal temperature difference between the external heat source and sink is rarely fully established across the thermoelectric legs. Instead, interfaces and packaging, hot-side heat collectors, cold-side heat sinks, and bypass heat leakage form a parasitic thermal-resistance chain that consumes a large fraction of the available temperature difference. This review develops a thermal-resistance-chain framework to describe heat-source access, interface/package losses, thermoelectric heat transport, cold-side rejection, and parasitic leakage under different thermal boundary conditions. Based on this framework, recent advances in module-level interfaces and packaging, hot-side heat collectors, and cold-side heat sinks are critically reviewed. The analysis shows that TEG thermal management should not be understood as isolated enhancement of a single component. Its central task is to identify the dominant resistance under a given heat-source boundary, quantify the temperature-loss budget, and coordinate the full thermal-resistance chain to maximize the effective working temperature difference. Current research is limited by scenario-dependent dominant resistance, the mismatch between local heat-transfer enhancement and net system gain, idealized steady-state evaluation, and the lack of unified reporting criteria. Future work should move toward full-chain co-design, with net power output, auxiliary energy consumption, temperature uniformity, transient response, and long-term reliability evaluated within a common framework. This perspective provides a system-level basis for translating component-level advances into robust and scalable TEG performance.

1. Introduction

1.1. Background

Improving the efficiency of existing energy systems is now a practical priority under rising energy demand and tightening decarbonization targets. Only about 30%–40% of the energy released from fossil fuel combustion is typically converted into useful work or electricity, while the remaining 60%–70% is rejected as waste heat [1]. This thermal energy is widely distributed in industrial processes, transportation systems, solar irradiation, geothermal environments, and even human metabolism [2]. Although the temperature range and heat flux vary greatly among these sources, their overall recovery potential is substantial [3].

Thermoelectric generators (TEGs) convert a temperature difference directly into electricity through the Seebeck effect, without requiring a working fluid or moving parts. Their compact structure, quiet operation, modularity, and reliability make them attractive for waste heat recovery and distributed power generation. Their feasibility has long been demonstrated in radioisotope thermoelectric generators for space missions [4], and recent advances in materials, fabrication, and packaging have expanded their use to industrial waste heat recovery [5],

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automotive exhaust utilization [1], especially in heavy-duty [6] and electrified trucks [7], solar [8] and geothermal power generation [9], and self-powered wearable and IoT devices [10].

Despite this broad potential, the practical output of TEG systems remains far below the level implied by material performance alone [11]. In most applications, the nominal temperature difference between the external heat source and sink cannot be fully established across the thermoelectric elements. The central issue is therefore not only how well the material converts heat into electricity, but how much of the available temperature difference can be preserved at the module and system levels.

1.2. From material advances to system bottlenecks

Thermoelectric research has long been driven by the improvement of intrinsic material performance, usually measured by the dimensionless figure of merit, ZT. Major progress has been achieved through band engineering, defect control, and multiscale structural design [12,13]. For example, Sn-doped p-type $AgSbTe_2$ has reached a peak ZT of about 2.5 and a unicouple conversion efficiency of 12.1% [14], while GeTe-based systems have also reported peak ZT values around 2.5 [15]. These results confirm that thermoelectric materials now possess strong intrinsic conversion capability.

For practical TEG systems, however, material improvement alone is insufficient. A high average ZT does not guarantee high device efficiency, because actual output is also constrained by thermal boundary conditions, geometry, electrical contact, thermal contact, and parasitic heat leakage [16]. In other words, material-level capability and system-level performance are not directly equivalent.

The origin of this gap lies in the parasitic thermal-resistance network that inevitably exists in practical TEGs. Heat must pass through the heat collector, interface layers, substrate, module body, and heat sink before it can be converted and rejected. Experimental studies show that the thermal contact resistance between the module and the heat source or heat sink can account for more than 80% of the total contact thermal resistance [17]. A distributed thermal/electrical contact resistance model further showed that treating these resistances as lumped parameters overestimates power output and efficiency by 16.9% and 24.5%, respectively [18], while an 80% reduction in thermal contact resistance can raise the maximum output power by 35.6%. These findings indicate that parasitic thermal resistance is not a secondary correction, but a primary factor governing temperature-difference utilization at the system level.

More importantly, the dominant resistance is not universal. In automotive exhaust recovery, industrial fluid waste heat, and high-temperature fuel-cell exhaust, the main limitations usually lie in hot-side heat transfer and cold-side heat rejection. In high-temperature solid waste heat and building wall applications, contact resistance and spreading resistance become more important. In wearable and bioheat systems, body-side conduction, imperfect skin contact, and parasitic heat leakage often dominate. In solar TEGs, optical heat input, hot-side thermal loss, and cold-side environmental conditions act together [19]. It has also been shown that, in solar thermoelectric systems, the dominant resistance may lie in cold-side convective heat transfer [20], whereas in passive geothermal systems, reducing the total exchanger thermal resistance by 26% can further increase the benefit of module geometry optimization by 60% [21]. The bottleneck in TEGs therefore shifts with heat-source type, structural configuration, and operating condition.

These observations indicate that system-level thermal management is the bridge between material performance and practical output. The key issue is not merely to reduce one local thermal resistance, but to determine how the available source-to-sink temperature difference is divided among heat-source access, interface/package losses, thermoelectric heat transport, cold-side rejection, and parasitic leakage. Therefore, TEG thermal management requires a framework that can identify the dominant resistance, locate temperature loss along the heat-transfer path, and guide the coordinated design of interfaces and packaging, hot-side heat collectors, and cold-side heat sinks, as illustrated in Fig. 1.

1.3. Scope of this review

Existing TEG reviews can be broadly grouped into several categories. Material-oriented reviews mainly focus on thermoelectric material systems, routes for improving ZT, and basic device physics [22–24]. Application-oriented reviews discuss specific scenarios such as wearable devices, solar/radiative systems, geothermal energy, and automotive waste heat recovery [25,26]. Other reviews emphasize structural design, additive manufacturing, topology optimization, system integration, or techno-economic optimization [27,28]. Table 1 positions representative reviews that are most relevant to TEG thermal management.

Among previous studies, the review by Fernández-Yáñez et al. [29] is the closest to the present topic. It focused on TEG thermal management for waste energy recovery and provided useful guidance on heat-source integration, heat-exchanger heat-transfer/pressure-drop trade-off, insulation, parasitic losses, net recovered energy, and system-efficiency evaluation. Building on this foundation, the present review shifts the organizing logic from heat-source-specific design and prototype evaluation to a thermal-resistance-chain perspective. The central question is how the nominal source-to-sink temperature difference is consumed by heat-source access, interface/package losses, thermoelectric heat transport, cold-side rejection, and parasitic leakage before it becomes the effective temperature difference across the thermoelectric legs.

From this perspective, several gaps remain in the existing literature. First, the nominal source-to-sink temperature difference and the effective temperature difference across the thermoelectric legs are not always clearly distinguished, although the latter directly determines electrical output. Second, hot-side heat access, interface/package resistance, cold-side heat rejection, and parasitic heat leakage are often treated as separate design issues, even though they compete for the same temperature-difference budget. Third, cross-scenario synthesis is still insufficient for explaining why the dominant resistance shifts among exhaust, solar, geothermal, combustion, industrial, building-integrated, and wearable TEG systems, and how this shift changes the priority of thermal-management strategies. Without such a framework, it remains difficult to judge whether a local heat-transfer enhancement can be translated into net system-level gain [29].

This review addresses these gaps by developing a thermal-resistance-chain framework for TEG thermal management. Its main contribution is not only a literature summary, but a system-level diagnostic framework for dominant-resistance identification, temperature-loss budgeting, net-power evaluation, and full-chain co-design. Based on this framework, recent advances are reviewed at three levels: interfaces and packaging, hot-side heat collectors, and cold-side heat sinks. The review further compares dominant-resistance shifts across representative application scenarios and discusses future directions in transient operation, durability assessment, auxiliary-loss evaluation, and standardized reporting. The aim is to provide a basis for translating component-level thermal-management advances into robust and scalable TEG system performance.

1.4. Review methodology

A structured literature search was conducted to assemble the corpus for this review. Web of Science, Scopus, ScienceDirect, and Google Scholar were searched, with primary attention given to publications from 2021 to 2026. Earlier studies were retained when they provided essential theoretical background, governing models, benchmark results, or widely cited design principles. The main search terms included “thermoelectric generator”, “TEG”, “thermal management”, “thermal

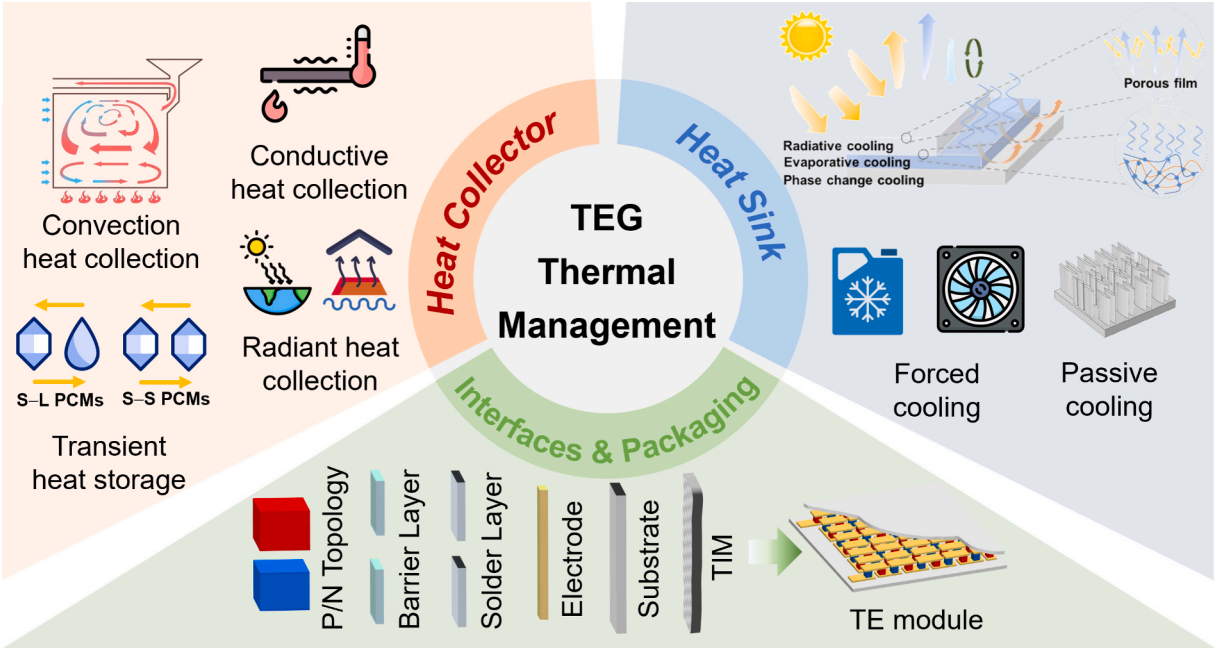

Fig. 1. Thermal management of TEGs.

Table 1
Positioning of this review relative to recent TEG review articles.

Review	Year	Coverage period	Main scope	Thermal-management focus	Gap relative to this review
Fernández-Yáñez et al. [29]	2021	2000s-2021	Thermal management of TEGs	Heat-source integration, heat exchangers, and application-related thermal-management strategies	Centered on waste-heat-source integration and prototype evaluation
Feng et al. [27]	2023	2010–2023	Environmental energy harvesting	Solar, radiative, ocean, and geothermal TEGs	No full thermal-resistance-chain framework
Li et al. [30]	2025	2010–2024	Micro-combustion-powered TEGs	Combustor–TEG integration and hot-side collection	Scenario-specific; limited cross-scenario generalization
Cui et al. [31]	2025	2011–2025	TEG principles and applications	Broad application and device overview	Thermal resistance not used as the central framework
Mousavi et al. [32]	2025	2000s-2025	Radiative-cooling TE systems	Radiative cold-side cooling	Focused on radiative cooling; hot-side and package losses are secondary
Siddique et al. [33]	2026	2000s-2026	Non-conventional TE leg geometry	Internal leg-shape optimization	External heat access, interfaces, and heat sinks are outside the main scope
Goswami et al. [34]	2026	2008–2025	TEG heat-recovery systems	System design, optimization, and techno-economics	Temperature-loss budgeting is not the organizing logic
This study	–	2021–2026	Cross-scenario TEG thermal management	Interfaces, hot-side collectors, cold-side sinks, and leakage	Builds a full resistance-chain framework for dominant-resistance diagnosis and co-design

resistance", "thermal contact resistance", "interface", "packaging", "substrate", "heat collector", "heat exchanger", and "heat sink". These terms were used individually and in combination to cover both component-level thermal management and application-specific TEG systems.

The screening process was first based on titles, abstracts, and keywords, followed by full-text assessment. Articles were included when they addressed heat-source access, interface/package resistance, thermoelectric heat transport, cold-side heat rejection, parasitic heat leakage, temperature-difference utilization, net power output, transient response, or thermal reliability. Studies focusing only on thermoelectric material synthesis or ZT improvement were excluded unless they contained device- or system-level thermal-management information. Thermoelectric-cooling studies were also excluded unless their heat-transfer structures or boundary-condition analysis were directly relevant to TEG operation.

Review articles were used to define the research context and to compare the present work with existing review scopes, as summarized in Table 1. Original research articles provided the main technical evidence for Sections 3–5. The representative studies in Table 3 were selected to cover different application scenarios, heat-source types, heat-collector structures, cold-side heat sink strategies, operating boundary conditions, and reported performance metrics. Priority was given to studies that reported sufficient thermal-boundary information, such as source/sink temperatures, module-side temperatures, flow or heat-flux conditions, cooling method, pressure drop, auxiliary energy consumption, gross or net output, and temperature uniformity. This selection supports cross-scenario comparison and helps identify the dominant thermal-resistance segments in different TEG applications.

2. Thermal resistance network model for TEGs

TEGs operate under thermal boundaries [35] that range from the small temperature difference between the human body and ambient air to the high heat fluxes encountered at industrial hot walls, combustion exhausts, and solar irradiation. Although these systems differ greatly in form, they are governed by the same question [36]: how much of the source-to-sink temperature difference can actually be preserved across the thermoelectric elements? A TEG should therefore be treated not as an abstract heat engine, but as a thermoelectric conduction process

constrained by external thermal boundaries [37]. This section establishes a generalized thermal resistance network for different heat-source conditions and uses it as the common framework for the later discussion of interface and packaging, hot-side heat collectors, and cold-side heat sinks. (See Fig. 2)

2.1. Generalized thermal resistance network

Let the far-field hot and cold source temperatures be $T_{\infty,h}$ and $T_{\infty,c}$, and the actual hot-side and cold-side temperatures of the thermoelectric legs be T_h and T_c, respectively. The nominal source-to-sink temperature difference is $\Delta T_\infty = T_{\infty,h}\text{-}T_{\infty,c}$, whereas the effective working temperature difference for power generation is $\Delta T = T_h\text{-}T_c$. To maintain a consistent description across different applications, all thermal obstacles outside the thermoelectric branch are grouped into an external thermal resistance network. Instead of treating them as a simple series chain, they are divided into four parts: hot-side source-access resistance, cold-side heat-rejection resistance, interface/package resistance, and parasitic heat leakage. This decomposition is necessary because source access and sink rejection often involve several parallel heat-transfer modes, and because hot-side and cold-side contact resistances do not affect performance symmetrically, with the cold-side contact resistance often being more detrimental [38].

The hot-side and cold-side external resistances, $R_{ext,h}$ and $R_{ext,c}$, are defined as

$$R_{ext,h} = R_{src,h} + R_{int,h} \tag{1}$$

$$R_{ext,c} = R_{int,c} + R_{sink,c} \tag{2}$$

where $R_{src,h}$ describes heat-source access from the far-field hot source to the module hot-side interface, $R_{sink,c}$ describes heat rejection from the cold-side heat sink to the far-field cold source, and $R_{int,h}$ and $R_{int,c}$ denote the hot-side and cold-side interface/package resistances, respectively.

For both sides, source access and heat rejection are first expressed through equivalent thermal conductance. On the hot side,

$$G_{src,h} = G_{conv,h} + G_{rad,h} + G_{cond,h} + G_{opt,h}, \quad R_{src,h} = G_{src,h}^{-1} \tag{3}$$

where $G_{conv,h}$, $G_{rad,h}$, $G_{cond,h}$, and $G_{opt,h}$ denote the equivalent thermal conductances associated with convection, radiation, bulk conduction, and optical/photothermal conversion, respectively. Their relative importance depends on the heat-source type. The cold side can be written in the same form as

$$G_{sink,c} = G_{conv,c} + G_{rad,c} + G_{liq,c} + G_{evap,c}, \quad R_{sink,c} = G_{sink,c}^{-1} \tag{4}$$

where the equivalent conductance may arise from air convection, radiative cooling, liquid cooling, or evaporative cooling.

Interface and package resistances are introduced in series on both sides:

$$R_{int,j} = R_{spread,j} + R_{contact,j} + R_{sub,j} + R_{attach,j}, \quad j = h, c \tag{5}$$

where $R_{spread,j}$ represents spreading resistance caused by size mismatch, $R_{contact,j}$ is solid-solid contact resistance, $R_{sub,j}$ is the resistance of the ceramic substrate or intermediate insulating layer, and $R_{attach,j}$ is the resistance of the adhesive, solder, or intermediate bonding layer. In addition to the main heat-transfer path, the module may also contain a bypass heat-leakage branch that directly links the hot side and cold side. Its equivalent thermal conductance is written as

$$K_{leak} = \frac{1}{R_{leak}} \tag{6}$$

This term accounts for parasitic heat leakage through fillers, package frames, and sidewall radiation or convection. When bypass leakage is negligible, $K_{leak} \to 0$, and the model reduces to the conventional three-part chain of hot-side external resistance, thermoelectric module, and cold-side external resistance. When leakage is significant, however, the system can no longer be treated as a simple series network [39].

2.2. Thermoelectric coupled governing equations

To provide a common basis for the following sections, a steady-state, one-dimensional, averaged-property model is adopted. Let α be the effective Seebeck coefficient, R_{elec} the effective electrical resistance, K_{TE} the open-circuit thermal conductance of the module, and $R_{TE} = 1/K_{TE}$ the corresponding open-circuit thermal resistance. Here, R_{elec} includes both the intrinsic electrical resistance of the thermoelectric legs and the electrical contact resistance. If the Thomson term is not written explicitly, the hot-side and cold-side heat fluxes of the thermoelectric branch are

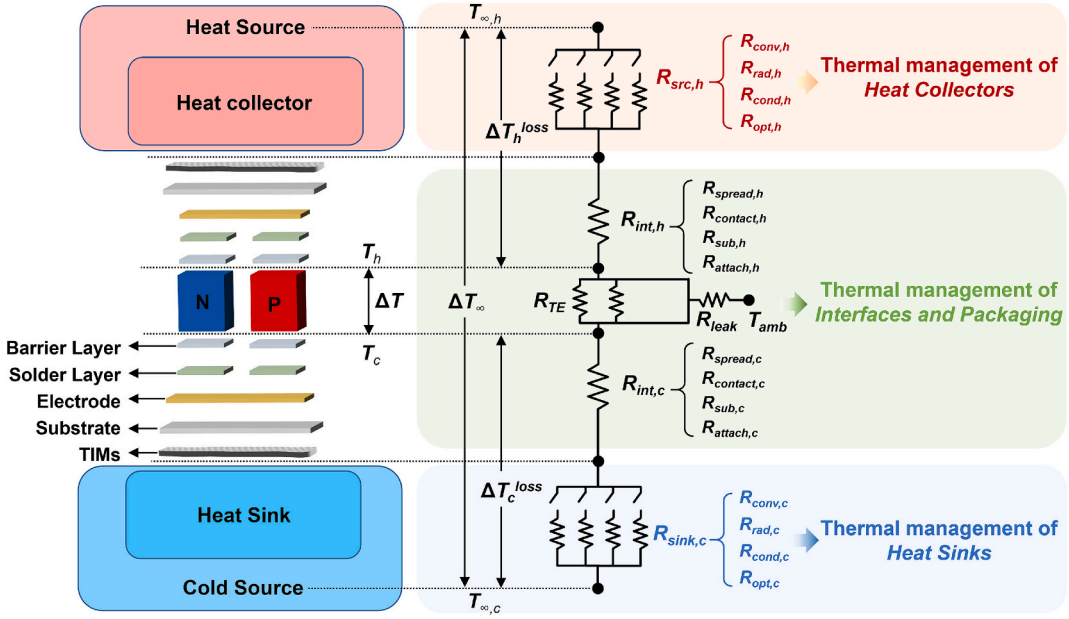

Fig. 2. Thermal resistance model.

$$Q_h = K_{TE}(T_h - T_c) + \alpha I T_h - \frac{1}{2} I^2 R_{elec} \tag{7}$$

$$Q_c = K_{TE}(T_h - T_c) + \alpha I T_c + \frac{1}{2} I^2 R_{elec} \tag{8}$$

where the first term represents Fourier conduction, the second Peltier heat, and the third Joule heat, assumed to be equally distributed to the two sides. The corresponding electrical relations are

$$V = \alpha(T_h - T_c) - I R_{elec} \tag{9}$$

$$I = \frac{\alpha(T_h - T_c)}{R_{elec} + R_L} \tag{10}$$

$$P = I^2 R_L = \frac{\alpha^2 (T_h - T_c)^2 R_L}{(R_{elec} + R_L)^2} \tag{11}$$

where R_L is the external load resistance.

Coupling the thermoelectric branch with the external thermal resistance network yields the hot-side and cold-side node balances:

$$\frac{T_{\infty,h} - T_h}{R_{ext,h}} = Q_h + K_{leak}(T_h - T_c) \tag{12}$$

$$\frac{T_c - T_{\infty,c}}{R_{ext,c}} = Q_c + K_{leak}(T_h - T_c) \tag{13}$$

Accordingly, the nominal source-to-sink temperature difference can be written in a more intuitive form as

$$\Delta T_\infty = \underbrace{(Q_h + K_{leak}\Delta T)R_{ext,h}}_{\Delta T_h^{loss}} + \Delta T + \underbrace{(Q_c + K_{leak}\Delta T)R_{ext,c}}_{\Delta T_c^{loss}} \tag{14}$$

This expression separates the thermal-management problem into three directly actionable targets: reducing the hot-side loss ΔT_h^{loss}, reducing the cold-side loss ΔT_c^{loss}, and suppressing the leakage term $K_{leak}\Delta T$. The constant-property model is suitable as a baseline for system-level resistance mapping. For large temperature spans, strong temperature dependence, or high-temperature material systems, α, ρ, and κ should be treated as temperature-dependent, and the Thomson term should be retained explicitly.

2.3. Effective working temperature difference, equivalent thermal resistance, and thermal matching

For a given thermoelectric system, the actual output is governed by ΔT, not by ΔT_∞. Under open-circuit conditions ($I = 0$), the Peltier and Joule terms vanish and the system reduces to a pure thermal-resistance divider. In this case, the thermoelectric branch and the leakage branch form an equivalent thermal conductance

$$K_0 = K_{TE} + K_{leak}, \quad R_0 = \frac{1}{K_0} \tag{15}$$

and the effective working temperature difference becomes

$$\Delta T_{oc} = \Delta T_\infty \frac{R_0}{R_{ext,h} + R_0 + R_{ext,c}} \tag{16}$$

Defining the temperature-utilization factor as $\phi_T = \Delta T/\Delta T_\infty$, the open-circuit result is

$$\phi_{T,oc} = \frac{R_0}{R_{ext,h} + R_0 + R_{ext,c}} \tag{17}$$

This relation shows that ϕ_T is reduced by two effects: the series external resistances on the hot and cold sides, which weaken thermal access to the module, and the bypass leakage path, which dissipates part of the available temperature difference before it can be used for power generation.

Under load, R_{TE} is no longer load-independent, because the hot-side and cold-side heat fluxes are no longer equal and Peltier heat transport changes the effective thermal conductance of the module. If the average heat flux $Q_m = (Q_h + Q_c)/2$ is used to represent the main heat flow through the module, an approximate relation can be written as

$$Q_m = \left(K_{TE} + \frac{\alpha^2 \overline{T}}{R_{elec} + R_L}\right)\Delta T, \quad \overline{T} = \frac{T_h + T_c}{2} \tag{18}$$

With parasitic heat leakage included, this becomes

$$K_{eff}(R_L) \approx K_{TE} + K_{leak} + \frac{\alpha^2 \overline{T}}{R_{elec} + R_L} \tag{19}$$

and correspondingly,

$$\Delta T(R_L) \approx \frac{\Delta T_\infty}{1 + (R_{ext,h} + R_{ext,c})K_{eff}(R_L)} \tag{20}$$

As the load decreases and the current increases, the effective thermal conductance rises, which further compresses ΔT. The effective working temperature difference under load is therefore usually smaller than that under open-circuit conditions. The implication is direct: the optimal load, geometry, and hot-side/cold-side structures cannot be determined independently of the external thermal resistances.

The resistance-chain framework has different levels of rigor for different heat-transfer modes. For steady conduction, (hA)-based convection, contact resistance, and spreading resistance, the resistance terms represent direct heat-rate/temperature-difference relations. For optical/photothermal input, radiation, combustion heating, geothermal replenishment, and transient thermal storage, they are used as locally linearized equivalent parameters around a specified operating state.

Here, $T_{\infty,h}$ denotes the far-field hot-source temperature for conventional fluid or solid heat sources. Under a constant-heat-flux boundary, it represents an equivalent source temperature associated with the imposed heat input. For solar, combustion, geothermal, or other environmental sources, it should be interpreted as the local effective source temperature at the collector boundary. $G_{opt,h}$ is not an intrinsic temperature-difference-driven conductance, but an equivalent optical/photothermal access parameter defined after the irradiance, absorber area, photothermal efficiency, and reference thermal potential are specified. Radiative terms are treated as linearized exchange coefficients evaluated around the operating temperature.

2.4. Dominant thermal resistances under different application scenarios

Based on the generalized model above, TEG thermal management can be reduced to four tasks: reducing the hot-side source-access resistance, reducing interface/package resistance, reducing the cold-side heat-rejection resistance, and suppressing parasitic heat leakage. As summarized in Table 2, the essential difference among application scenarios lies in which segment of the thermal resistance chain becomes dominant.

For convection-limited systems, such as engine exhaust recovery and industrial fluid pipelines, both the hot side and the cold side are often close to Robin-type boundaries, with the dominant terms given by $R_{src,h} \sim 1/(h_h A_h)$ and $R_{sink,c} \sim 1/(h_c A_c)$. In these cases, performance improvement mainly depends on enlarging the effective heat-transfer area, optimizing flow distribution, and increasing the effective hA under pressure-drop or pumping-power constraints.

For solid-contact- and spreading-limited systems, such as high-temperature furnace walls, industrial solid waste heat, and molten slag, heat transfer is primarily constrained by $R_{contact,h}$ and $R_{spread,h}$. Here, the key issue is transferring heat more effectively into the module hot side through compliant interlayers, surface metallization, thermal spreaders, and stable contact pressure.

For bulk-conduction- and nonconformal-contact-limited systems, including wearables, bioheat harvesting, shallow geothermal sources,

Table 2
Thermal resistance importance in representative TEG application scenarios.

Application scenario	$R_{conv,h}$	$R_{rad,h}$	$R_{cond,h}$	$R_{opt,h}$	$R_{int,h}$	R_{leak}	$R_{int,c}$	R_{sink}	Primary bottleneck(s)
Automotive exhaust	Dominant	Weak	Absent	Absent	Secondary	Secondary	Secondary	Important	Hot-side convection; cold-side cooling under net-power constraint
Industrial fluid/ exhaust	Dominant	Weak	Absent	Absent	Secondary	Secondary	Secondary	Important	Hot-side convection; flow maldistribution; sink-side resistance
Industrial solid/ radiative heat	Secondary	Important	Important	Absent	Important	Secondary	Secondary	Important	Contact/spreading resistance; radiative or non-contact heat access
Solar/radiative TEGs	Weak	Important	Absent	Dominant	Secondary	Important	Secondary	Dominant	Optical/photothermal access; cold-side cooling; hot-side thermal loss
Geothermal	Secondary	Absent	Dominant	Absent	Important	Secondary	Secondary	Important	Source-side conduction path; exchanger resistance; environmental cooling
Wearable and bioheat systems	Absent	Weak	Dominant	Absent	Dominant	Important	Weak	Secondary	Body-side conduction; conformal contact; parasitic heat leakage
Combustion / biomass / stove systems	Important	Important	Secondary	Absent	Important	Secondary	Secondary	Important	Mixed radiative/convective heat input; hot-side spreading; fouling-related resistance
Building heat	Weak	Secondary	Dominant	Weak	Important	Important	Secondary	Secondary	Long conduction path; low-grade temperature difference; structural heat leakage

Note: The labels reflect the typical importance of each resistance term under a given thermal boundary: Dominant, Important, Secondary, Weak, and Absent indicate decreasing relevance. The classification is based on reported trends and literature synthesis, and may vary with geometry, heat flux, contact condition, cooling method, and operating environment.

soil heat sources, and some building envelopes, the dominant terms usually shift to $R_{cond,h}$ and $R_{contact,h}$, while parasitic heat leakage K_{leak} also becomes more important. In these cases, thermal management should focus on improving conformal contact, shortening the heat-transfer path, and suppressing lateral heat leakage, rather than simply enhancing local heat transfer.

For radiation- or photothermal-access-limited systems, such as solar TEGs and radiative-coupled systems, the dominant hot-side term is mainly $R_{opt,h}$. The governing competition is among solar absorption, photothermal conversion, and hot-side thermal loss. If radiative cooling is also adopted on the cold side, $R_{sink,c}$ becomes partly controlled by radiative boundary conditions. In these systems, performance improvement relies on reducing optical access resistance and suppressing undesired radiative and convective losses from the hot side.

Overall, the generalized thermal resistance network provides a common framework for the rest of this review. Section 3 focuses on interfaces, packaging, and parasitic leakage, with emphasis on $R_{int,h}$, $R_{int,c}$, and K_{leak}. Section 4 addresses hot-side heat collectors, with emphasis on reducing $R_{src,h}$ and improving hot-side heat spreading. Section 5 addresses cold-side heat sinks, with emphasis on reducing $R_{sink,c}$ and maintaining cold-side temperature uniformity.

3. Thermal management of interfaces and packaging

In practical TEG systems, only part of the nominal source-to-sink temperature difference is imposed across the thermoelectric legs. Before entering the active thermoelectric materials, heat must pass through external thermal interfaces, ceramic or flexible substrates, metallized layers, bonding layers, diffusion barriers, electrodes, fillers, encapsulants, and geometric contact regions. These components introduce contact resistance, spreading resistance, through-plane conduction resistance, lateral heat leakage, and thermomechanical degradation. Therefore, interfaces and packaging should not be treated as passive structural accessories, but as a module-level thermal-resistance network that directly controls the effective working temperature difference.

The central task of package-level thermal management is not to minimize every local resistance independently. A highly conductive layer may still perform poorly if it is too thick, too stiff, poorly bonded, or unstable under thermal cycling. Conversely, a low-conductivity region can be beneficial if it suppresses lateral bypass heat leakage. Thus, the relevant design criterion is resistance allocation: heat should be guided preferentially through the thermoelectric legs while parasitic temperature losses, electrical contact losses, and mechanical damage are jointly constrained [40]. This section reviews four package-level design domains: thermal interface materials, electrodes and interconnects, substrates, and thermoelectric-element geometry.

3.1. Thermal interface materials (TIMs)

The external assembly interfaces between the heat exchanger and the module substrate are often among the largest sources of temperature-difference loss in integrated TEG systems [41]. These interfaces are particularly critical when the heat source is rough, curved, mechanically constrained, or operated under limited clamping pressure. The role of thermal interface materials is therefore to reduce constriction resistance by increasing real contact area, filling micro-gaps, and maintaining a stable heat-transfer path across the hot-side and cold-side contacts.

A consistent conclusion from recent studies is that TIM performance is governed less by nominal thermal conductivity than by the coupled effects of compliance, bond-line thickness, pressure dependence, and long-term stability [42]. Gu et al. [43] showed that thermal grease can outperform graphite paper despite its lower bulk thermal conductivity because it conforms more effectively to rough surfaces. Li et al. [44] further identified a cold-state contact pressure of about 1.2 MPa as a critical assembly threshold; below this value, insufficient contact reduced output by 11.7%, whereas further pressure increase produced only marginal gain. These results indicate that TIM design is fundamentally a pressure-dependent interfacial problem rather than a simple high-conductivity material selection problem.

Recent TIMs have therefore shifted toward compliant composite interfaces. Liquid-metal and particle-filled elastomers are representative examples because they combine moderate-to-high thermal conductivity with low modulus and surface conformity [45]. Zhang et al. [46] reported a Ga-based liquid-metal/Cu elastomeric TIM with a thermal conductivity of 3.94 $W\,m^{-1}\,K^{-1}$ and a Young's modulus of only 699 kPa, where the deformable liquid-metal phase bridges conductive particles under compression, as shown in Fig. 3(a) and (b). Such materials reduce both intrinsic conduction resistance and contact constriction resistance. However, their system-level value still depends on leakage suppression, oxidation resistance, compatibility with adjacent electrodes and substrates, and cycling stability.

TIMs are also evolving from passive gap fillers to multifunctional interfacial layers. In PV-TEG systems, pyrolytic graphite sheets have been used to improve overall efficiency relative to conventional thermal grease [48]. Liquid-metal-based photothermal interfaces further integrate light absorption, adhesion, and heat conduction [49]. Fu et al. [47]

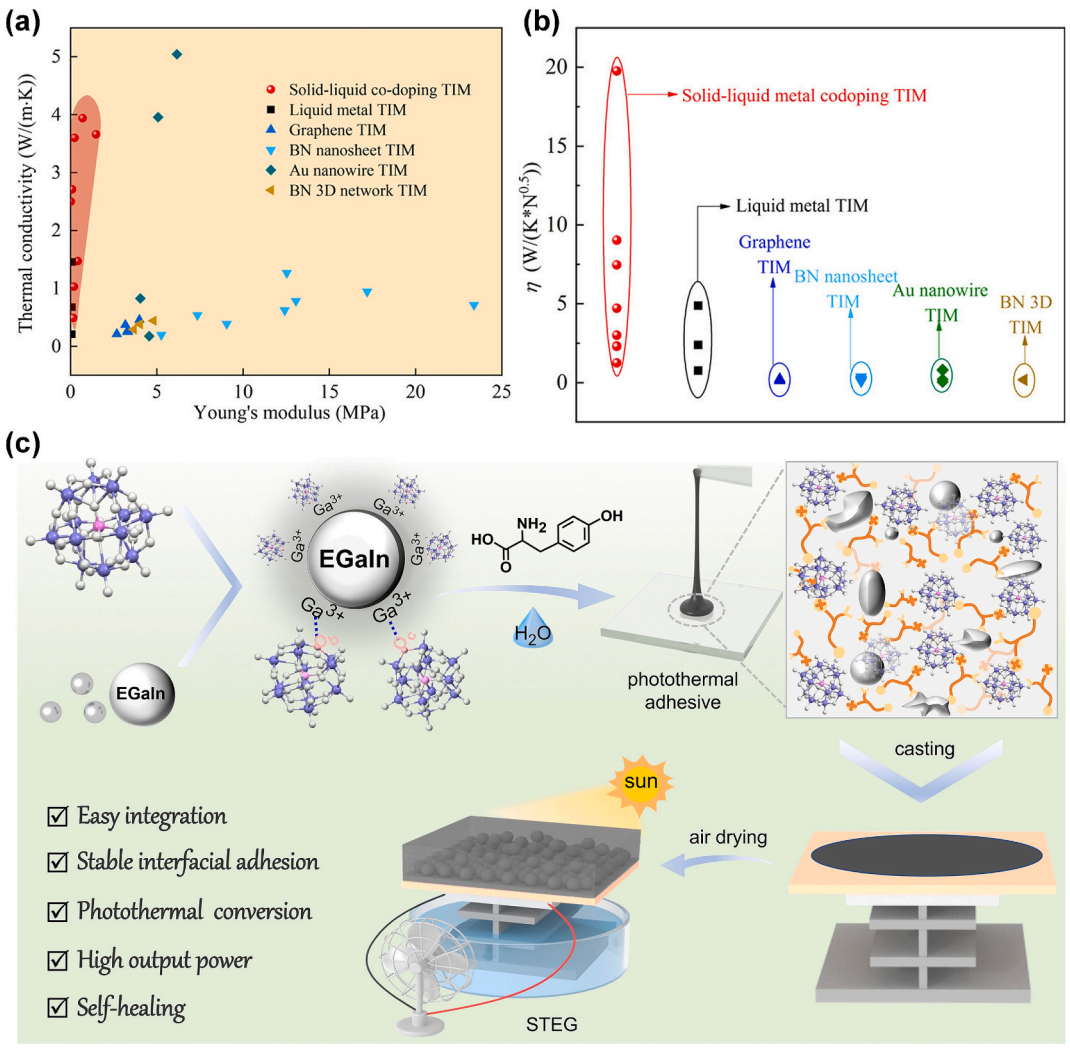

Fig. 3. Recent advances in TIMs. (a) Comparison of solid–liquid metal co-doped TIMs with other TIMs in thermal conductivity and Young's modulus. (b) Comparison of comprehensive thermomechanical performance factors of different TIMs. Reproduced by permission [46]. Copyright 2023 American Chemical Society. (c) Heteropoly blue/liquid metal supramolecular adhesive coating with photothermal conversion and self-healing capability. Reproduced by permission [47]. Copyright 2026 Elsevier. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

developed a heteropoly blue/liquid-metal supramolecular adhesive coating that simultaneously fills interfacial voids and acts as a photothermal absorber, as shown in Fig. 3(c). These designs are attractive for solar TEGs, but their effects must be interpreted carefully because enhanced output may arise from both reduced contact resistance and increased optical heat input. Future TIM studies should therefore report contact thermal resistance under defined pressure, bond-line thickness, cycling stability, and actual temperature-difference utilization, rather than relying only on bulk thermal conductivity.

3.2. *Electrodes and interconnects*

Electrodes, bonding layers, diffusion barriers, and interconnects determine how heat and current enter the thermoelectric legs. Their function is not limited to reducing electrical contact loss [50]. They also govern local temperature drop, residual stress, interfacial reaction, and the long-term stability of heat flow [51]. In the thermal-resistance chain, these layers form an internal package bottleneck: even when external heat exchangers and TIMs are well designed, unstable joints or barriers can still consume the available temperature difference and accelerate performance decay [52].

Bonding layers represent the first critical interface. Conventional modules often use Cu electrodes and Sn-based solders [53], but thick solder layers add thermal resistance, while thermal-expansion mismatch among Cu, solder, and thermoelectric materials promotes cracking and delamination during cycling [54]. Low-temperature, low-pressure sintering and weak-solder strategies have therefore become important [55]. Ag nanoparticle bonding can decouple joining temperature from service temperature and reduce joining-induced stress [56], while Cu nanoparticle pastes can lower interfacial thermal loss without sacrificing joint strength [57]. The key issue is not only initial resistance reduction, but whether the joint can maintain low thermal and electrical resistance under repeated thermal stress.

Barrier layers introduce a second design trade-off. They must suppress harmful interdiffusion while preserving joining activity, wetting, adhesion, and low interfacial resistance. The ideal barrier layer should exhibit fast joining kinetics during fabrication but slow degradation kinetics during service [50,58], as illustrated in Fig. 4(c) and (d). In Bi_2Te_3 modules, porous Ti barriers relieve stress through low modulus and particle sliding while forming a stable $TiTe_2$ reaction layer for low-loss contact [59]. Atomic-scale studies of Ni/Bi_2Te_3 further show that a completely reaction-free interface is not always necessary [60]. The more important criterion is controlled reaction: a stable and conductive interfacial phase may be beneficial, whereas continuous reaction produces progressive resistance build-up.

This explains why multilayer barriers such as NiMo [61], Co-P/Ni-P

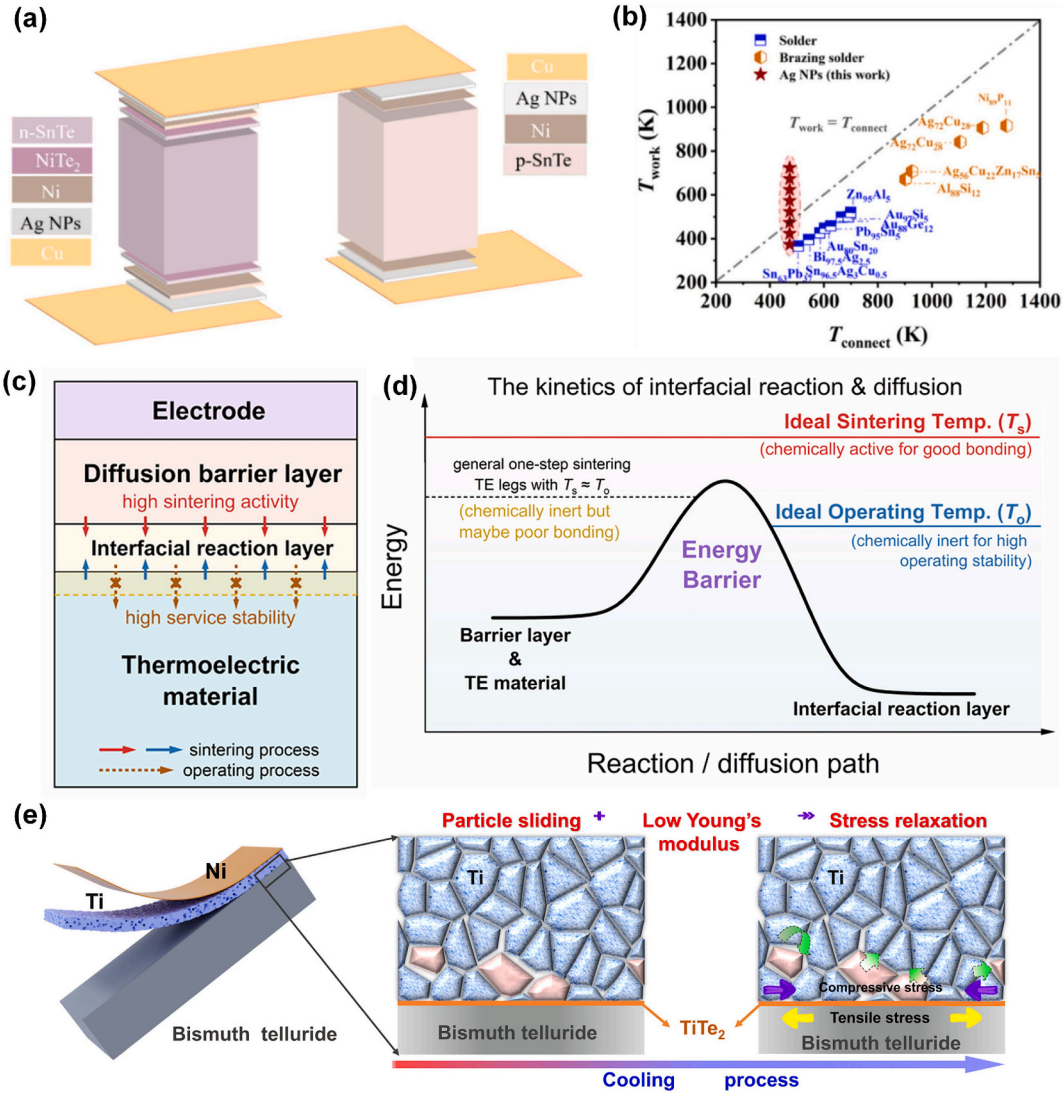

Fig. 4. Thermal management of electrode layers, diffusion barriers, and bonding layers in TEGs. (a) Ag nanoparticle bonding layer, with Ni used as the barrier for p-SnTe and $NiTe_2$ for n-SnTe. (b) Relationship between joining temperature and maximum device operating temperature for different materials. Reproduced by permission [55]. Copyright 2026 Elsevier. (c) Schematic of an ideal thermoelectric barrier layer with high joining activity and high service reliability. (d) Barrier-layer design based on the kinetic contrast between joining and operation. Reproduced under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [50]. Copyright 2025 The Authors. (e) Working mechanism of the porous Ti barrier layer, where particle sliding and low Young's modulus release interfacial stress during cooling. Reproduced under a Creative Commons Attribution 4.0 International License [59]. Copyright 2023 The Authors.

[62], and Ni/Ti [63] are often more effective than single-metal barriers. They allow diffusion suppression, wetting, adhesion, thermal-expansion matching, and stress relaxation to be distributed across different sub-layers. In flexible and thin-film TEGs, electrodes additionally serve as adhesion layers, barriers, stress buffers, and heat-spreading structures. Cu—Ni textile electrodes [64] and nanoscale Ti interlayers at PI/Cu and Cu/thermoelectric-film interfaces [65] illustrate this multifunctional role. However, high-conductivity electrodes can also create lateral thermal shortcuts if their geometry is not properly controlled. Future electrode studies should therefore report thermal contact resistance, electrical contact resistivity, joint thickness, cycling stability, and interfacial microstructural evolution together, because these quantities jointly determine whether an interconnect improves system-level temperature-difference utilization.

3.3. Substrates

Substrates provide electrical insulation, mechanical support, and heat-transfer regulation. Their role differs sharply between rigid modules and flexible/wearable devices. In rigid modules, substrates should reduce through-plane conduction resistance and spread heat uniformly before it enters the legs, while avoiding excessive thermal stress at ceramic-metal and metal-leg junctions [66]. In flexible devices, substrates must also regulate body-heat input, suppress lateral heat leakage, and maintain conformal contact under low pressure [67]. Thus, substrate design is not a universal high-conductivity problem, but a scenario-dependent resistance-allocation problem.

For rigid TEG modules, Al_2O_3, AlN, MgO, and SiC are commonly used because they combine electrical insulation with acceptable thermal conduction. However, substrate optimization cannot be reduced to ceramic thermal conductivity alone. Fagehi [68] showed that substrate thickness and leg length must be considered together, with AlN approaching the best performance under thin-substrate and short-leg conditions. More recent work has shifted toward thermomechanical co-design. Sun et al. [69] found that stress concentrates at metal interconnect–ceramic and metal interconnect-leg junctions, and proposed a stepped ceramic plate with blind holes that increased minimum module lifetime by 68% (Fig. 5 (a)). Yan et al. [70] identified substrate

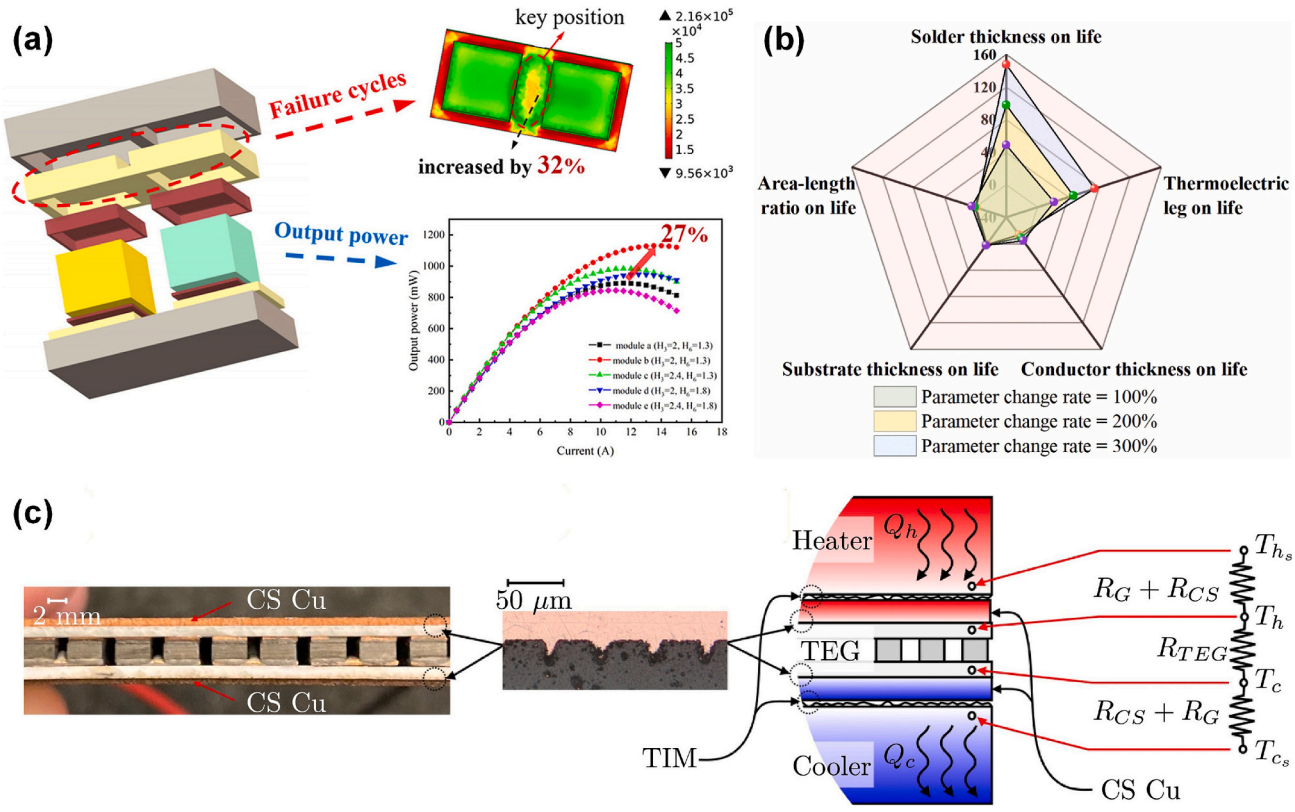

Fig. 5. Progress in insulating substrates for thermoelectric modules. (a) Stepped ceramic plate structure with blind holes. Reproduced by permission [69]. Copyright 2021 Elsevier. (b) Influence of thermoelectric element geometric parameters on lifetime. Reproduced by permission [70]. Copyright 2025 Elsevier. (c) Optical microscopy image of the cold-sprayed Cu/Al_2O_3 interface and the corresponding heat-flow network in the TEG module. Reproduced by permission [72]. Copyright 2024 Elsevier.

thickness, solder thickness, conductor thickness, and leg length as key lifetime parameters (Fig. 5 (b)), while Liu et al. [71] showed that mechanical constraints and electrode structures strongly affect stress distribution in high-temperature modules. These studies indicate that rigid-substrate thermal management must be evaluated together with package reliability.

Once bulk ceramic resistance is reduced, the ceramic–metal interface can become the next bottleneck. Tregenza et al. [73] used laser-textured Al_2O_3 combined with cold-sprayed Cu to form a low-resistance, high-adhesion bilayer. Their subsequent work showed that the maximum power loss of the cold-spray thermal interface was only 0–0.9%, compared with 0–9.9% for conventional mechanical-contact interfaces [72] (Fig. 5 (c)). This result illustrates a typical bottleneck transfer in the thermal-resistance chain: after the substrate material is improved, metallization quality, adhesion, and contact morphology may become equally important as the ceramic itself.

In flexible and wearable TEGs, the substrate is more than a support layer. It determines heat-input efficiency, lateral heat leakage, mechanical compliance, and wearing comfort. Miao et al. [74] showed that the thermal conductivity and Young's modulus of encapsulation materials must be co-optimized with leg geometry, ambient temperature, bending radius, and wearing pressure (Fig. 6 (a)) [75]. Ramesh et al. [76] used an aerogel-polydimethylsiloxane (PDMS) composite to reduce silicone thermal conductivity by about 50% and achieved nearly 35 μW cm^{-2} on the wrist. Ren et al. [77] further showed that heat leakage through the insulating substrate can reduce output by about one order of magnitude. These findings reveal that high thermal conductivity is not always beneficial in flexible devices; excessive in-plane conduction may bypass the thermoelectric legs.

Flexible substrates have therefore evolved toward anisotropic and layered heat-flow architectures. Kuang et al. [78] proposed a low-resistance flexible printed circuit board (FPCB) structure with openings and vias to promote directional heat conduction (Fig. 6 (b)-(c)). Huang et al. [81] combined a PI substrate, high-conductivity silica-AlN composite, and silica aerogel insulation to guide heat through the active legs. Zhao et al. [82] embedded bottom electrodes into an Al_2O_3/BN/PDMS composite substrate, turning the substrate into an active thermal pathway. Zhu et al. [79] further integrated a low-conductivity filler, liquid-metal/expanded-graphite photothermal substrate, and h-BN radiative-cooling layer, so that the substrate simultaneously controlled conduction, photothermal heating, and radiation (Fig. 6 (d)-(f)). These designs show that flexible substrates are moving from passive support to integrated heat-flow programming.

A further constraint is contact pressure. Wearable TEGs cannot rely on high clamping force, because comfort and skin safety limit allowable pressure [83,84]. Miao et al. [74] reported that 0.5–0.8 kPa is close to the perceptual limit for tight wear. Therefore, low-pressure conformity becomes as important as thermal conductivity. Semi-embedded legs with exposed electrodes (Fig. 6 (g)) [80], patterned high−/low-conductivity regions [85], staggered-gap designs [86], and three-dimensional thermoelectric textiles [87] all aim to preserve contact while controlling heat leakage. Future flexible-substrate studies should report contact pressure, bending radius, air permeability, thermal comfort, cycling durability, and the fraction of heat flowing through the thermoelectric legs. Otherwise, performance measured on flat laboratory fixtures may overestimate practical wearable output.

3.4. Geometric design of thermoelectric elements

Commercial TEG modules are typically based on rigid planar π-type architectures with constant-cross-section rectangular legs [88]. These designs are suitable for flat and stable heat sources, but become less

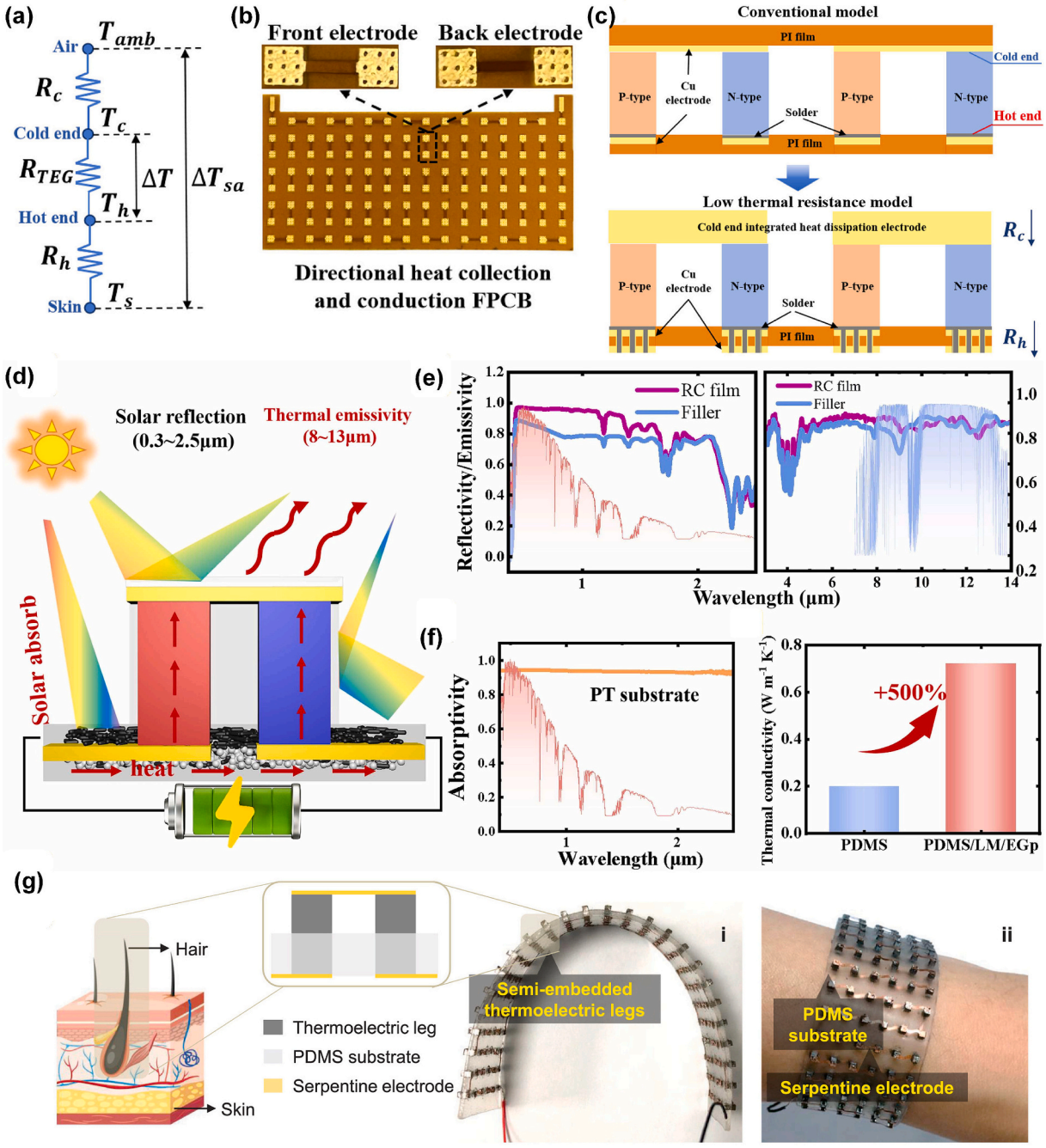

Fig. 6. Progress in flexible substrates. (a) Thermal resistance network of a wearable TEG. Reproduced under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [74]. Copyright 2024 The authors. (b) Schematic of directional heat collection and conduction using FPCB. (c) Comparison between conventional and low-resistance substrate configurations. Reproduced by permission [78]. Copyright 2025 Elsevier. (d) Thermal management mechanisms of PT substrate and RC film substrate. (e) UV reflectance and infrared emittance of the RC film. (f) UV absorption and thermal conductivity of the PT substrate. Reproduced by permission [79]. Copyright 2025 Elsevier. (g) Biomimetic FTEG with semi-embedded thermoelectric legs and exposed electrodes. Reproduced by permission [80]. Copyright 2025 Elsevier.

effective on curved, irregular, or deformable surfaces such as pipes, exhaust ducts, industrial walls, and human skin [89]. In these cases, incomplete geometric contact increases contact resistance and often requires thicker TIMs or auxiliary fixtures. Geometric design is therefore a thermal-management strategy because it changes contact conformity, heat-flow distribution, internal resistance, and thermomechanical stress [90].

For cylindrical and high-curvature heat sources, annular and tubular TEGs provide direct conformal matching [91]. Their main advantage is the reduction of curvature-induced contact loss and improvement of circumferential heat-flow uniformity (Fig. 7 (a)) [92]. Studies have reported improved efficiency under cylindrical heating, especially at higher inlet temperatures or stronger convective conditions [93,94]. Further gains have been pursued through variable cross-sections [88], asymmetric P/N leg ratios [92], segmented configurations [95], and optimized annular module layouts (Fig. 7(b)) [96,97]. However, annular structures are most justified only when curvature-induced contact resistance is the dominant bottleneck. If the system is limited by gas-side convection, cold-side rejection, or internal electrical resistance, curvature matching alone may yield limited net benefit.

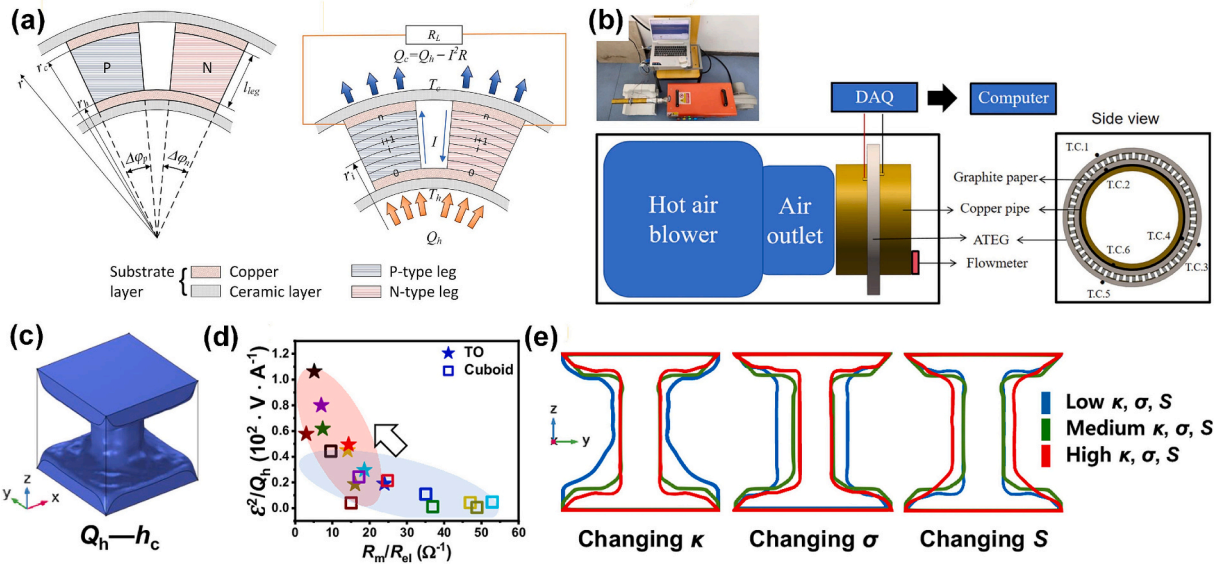

Fig. 7. Topology optimization of thermoelectric elements. (a) A pair of typical annular thermoelectric leg model. Reproduced by permission [92]. Copyright 2023 Elsevier. (b) An annular thermoelectric module applied to a cylindrical heat source. Reproduced by permission [97]. Copyright 2024 Elsevier. (c) Topology-optimized geometric configurations. (d) Efficiency enhancement of optimized topologies. (e) Optimization of heat-flow distribution through coupled topology and transport-property design. Reproduced under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [98]. Copyright 2026 The authors.

Manufacturing complexity, nonuniform stress, and nonstandard interconnection also remain important barriers.

Beyond module-level conformity, leg geometry has evolved from uniform rectangular prisms to variable-cross-section [99], asymmetric [100], and topology-optimized forms [101]. These designs redistribute thermal and electrical resistance within the active element [102]. Narrower or longer regions can help preserve a larger temperature difference, but they may also increase electrical resistance [103]. Larger contact areas can reduce local heat-transfer resistance, but may intensify heat leakage or stress concentration [104]. Therefore, geometric optimization should be compared under consistent constraints such as fixed material volume, footprint, boundary temperature, heat flux, and contact resistance [105]. Otherwise, reported gains may reflect more favorable boundary conditions rather than intrinsic geometric superiority.

Topology optimization provides a more systematic route because it can incorporate boundary conditions, interfacial resistance, package constraints, and mechanical stress into one design framework [98]. As shown in Fig. 7(c)-(e), optimized I-shaped or asymmetric hourglass-like structures can improve performance for specific materials and thermal boundaries. Its strength lies in treating geometry as a field variable rather than a set of predefined shapes. Its limitation lies in manufacturability, bonding feasibility, tolerance control, and long-term reliability. Thus, topology-optimized TEGs should be assessed not only by simulated efficiency enhancement, but also by contact resistance, fabrication yield, cycling stability, and package compatibility.

For wearable systems, geometry serves a different purpose. The heat source is soft, curved, and dynamically deformed, so fixed-curvature modules are insufficient. Island–bridge layouts, serpentine interconnects, kirigami/origami-inspired structures, and three-dimensional soft architectures decouple rigid thermoelectric legs from deformable electrical connections [106,107]. These designs improve real contact area under low normal pressure and reduce interfacial thermal loss. Soft thermal guide layers, soft insulating layers [108], three-dimensional architectures [109], and pop-up kirigami structures [110] can further sustain temperature difference while allowing bending or stretching. However, the added flexibility may introduce longer electrical paths, larger inactive areas, and additional soft-layer thermal resistance. Their value should therefore be judged by sustained power under realistic deformation rather than by flexibility alone.

Cellular and porous geometries, including honeycomb, microlattice, and textile-like structures, provide another route for resistance allocation [111]. By introducing pore networks, they can increase effective thermal resistance, reduce lateral heat leakage, and improve mechanical compliance [112]. These structures are particularly relevant to low-grade or wearable heat sources, where maintaining a small but stable temperature difference can be more important than maximizing heat throughput. Their limitation is that increased porosity may reduce absolute heat input, slow transient response, and lower power density [113]. Future studies should distinguish whether the observed improvement originates from better contact, reduced lateral leakage, higher effective thermal resistance, or mechanical compliance. Without this distinction, geometric design can easily be overinterpreted as a universal enhancement strategy.

A common lesson is that local thermal enhancement does not automatically produce system-level gain [114]. A high-conductivity TIM may fail under low pressure, a thin bonding layer may increase stress, a conductive substrate may create lateral leakage, and an optimized leg geometry may raise electrical resistance or manufacturing difficulty. Package-level thermal management should therefore be evaluated through contact thermal resistance, electrical contact loss, assembly pressure, layer thickness, temperature-difference utilization, cycling durability, and net output. From the thermal-resistance-chain perspective, the best package is not the one that minimizes every individual resistance, but the one that coordinates conduction, insulation, contact compliance, and mechanical stability to preserve the largest useful temperature difference over the operating lifetime.

4. Thermal management of heat collectors

The hot-side heat collector determines how external heat is converted into a usable thermal boundary for the thermoelectric module. In the thermal-resistance-chain framework, it mainly controls the hot-side source-access resistance, including convective, radiative, conductive, optical, and spreading components. The dominant design constraint depends strongly on the heat-source boundary. Exhaust-gas systems are limited by convective heat transfer and pressure drop; solar systems by

optical absorption, radiative/convective loss, and temporal intermittency; geothermal systems by long conduction paths and geological thermal replenishment; combustion-driven systems by mixed radiation–conduction–convection boundaries and fouling; industrial systems by spatially nonuniform heat fields and non-interference requirements; and building systems by low temperature gradients and architectural constraints. Therefore, heat collectors should be evaluated by their ability to reduce the dominant source-access resistance under the real boundary condition, rather than by local heat-transfer enhancement alone.

4.1. Automotive TEG heat collectors

Automotive TEG (ATEG) collectors must extract enthalpy from exhaust gas while avoiding excessive engine backpressure. Their central trade-off is therefore between increasing the gas-side heat-transfer coefficient and limiting aerodynamic loss [115] under different inlet-flow conditions [116]. Fins [117], twist tapes [118], perforated plates [119], metal foams [120], and microchannels [121] can enlarge the effective heat-transfer area and disturb the boundary layer, but the same structures may increase pressure drop and reduce net power [122]. This explains why recent ATEG studies have shifted from maximizing heat transfer to optimizing the combined heat-transfer and flow-resistance balance. Luo et al. [117] showed that fin spacing directly controls this balance, and optimized spacing increased net power and net efficiency by 9.52% and 22.95%, respectively.

More refined internal-flow designs attempt to redistribute exhaust flow without imposing excessive pressure penalty. Variable-twist tapes [118], perforated plates (Fig. 8(a)) [119,123], split fins [124], and vapor chambers [125] have been reported to improve output and voltage uniformity. Axially graded metal foams further reduce unnecessary upstream resistance and can decrease pressure drop by 35.9% with only a limited power penalty [126]. These results indicate that the relevant optimization variable is not simply hA, but hA per unit pressure drop, or more directly, net electrical gain per aerodynamic penalty.

A second bottleneck is streamwise temperature decay. Exhaust gas cools rapidly along the collector, so downstream modules experience lower hot-side temperatures and weaker temperature differences [130]. This causes module mismatch even if the inlet heat transfer is strong. Heat pipes, vapor chambers, and intermediate fluids have therefore been introduced to convert a nonuniform gas stream into a more uniform solid-side thermal boundary [131,132]. Luo et al. [127,133] integrated heat pipes (Fig. 8(b)) into ATEGs and reported power gains under both steady and transient conditions. Annular ATEGs coupled with vapor chambers achieved high temperature uniformity and net power output [134], while intermediate-fluid (Fig. 8(c)) systems use evaporation–condensation transport to improve output and reduce pressure drop simultaneously [128]. These structures do not merely enhance heat transfer; they reshape the hot-side boundary seen by the module.

Real driving cycles introduce a further constraint. Exhaust temperature and mass flow vary rapidly during acceleration, idling, and start–stop operation, making steady-state bench optimization insufficient [135,136]. Platform-specific co-design of heat-exchanger geometry and module arrangement [137] under constraints of mass, volume,

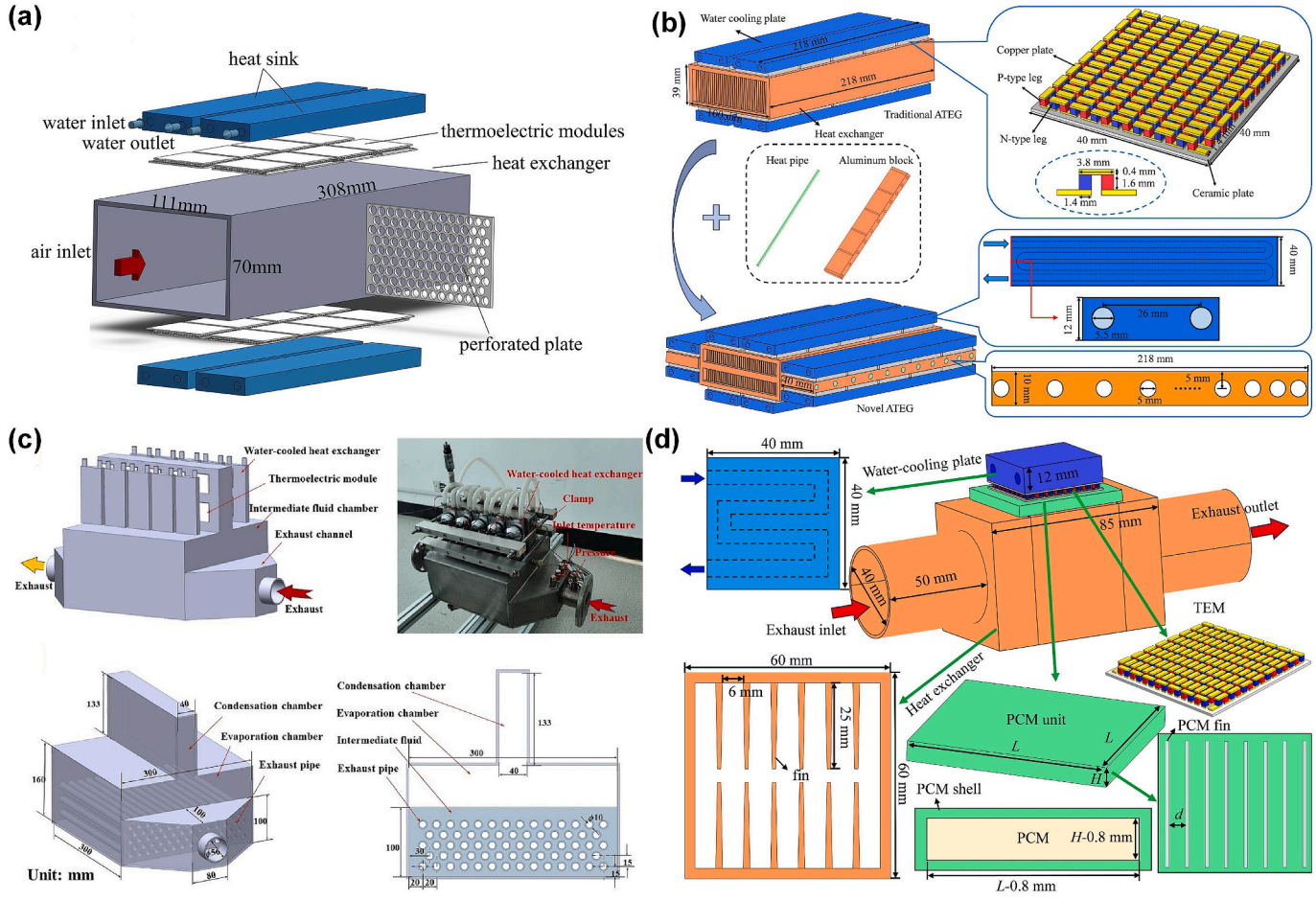

Fig. 8. Automotive TEG heat collectors. (a) Perforated-plate exhaust collector. Reproduced by permission [119]. Copyright 2023 Elsevier. (b) Fin–heat-pipe exhaust collector. Reproduced by permission [127]. Copyright 2025 Elsevier. (c) Intermediate-fluid collector. Reproduced by permission [128]. Copyright 2025 Elsevier. (d) PCM-assisted exhaust collector. Reproduced by permission [129]. Copyright 2026 Elsevier.

and pressure drop has been shown to improve net power and fuel efficiency in range-extended electric vehicles [138]. PCM-assisted collectors (Fig. 8(d)) provide another route by buffering exhaust fluctuations [129,139]. Although PCMs may reduce instantaneous peak output because of increased thermal inertia, they can suppress power fluctuations and protect modules from thermal shock [140].

Overall, ATEG collectors are governed by a three-way constraint among heat-transfer enhancement, aerodynamic penalty, and transient thermal matching. Future studies should report pressure drop, engine backpressure, net electrical output, temperature uniformity, and performance over driving cycles. Without these metrics, a collector that appears effective under steady exhaust flow may not produce a real vehicle-level gain.

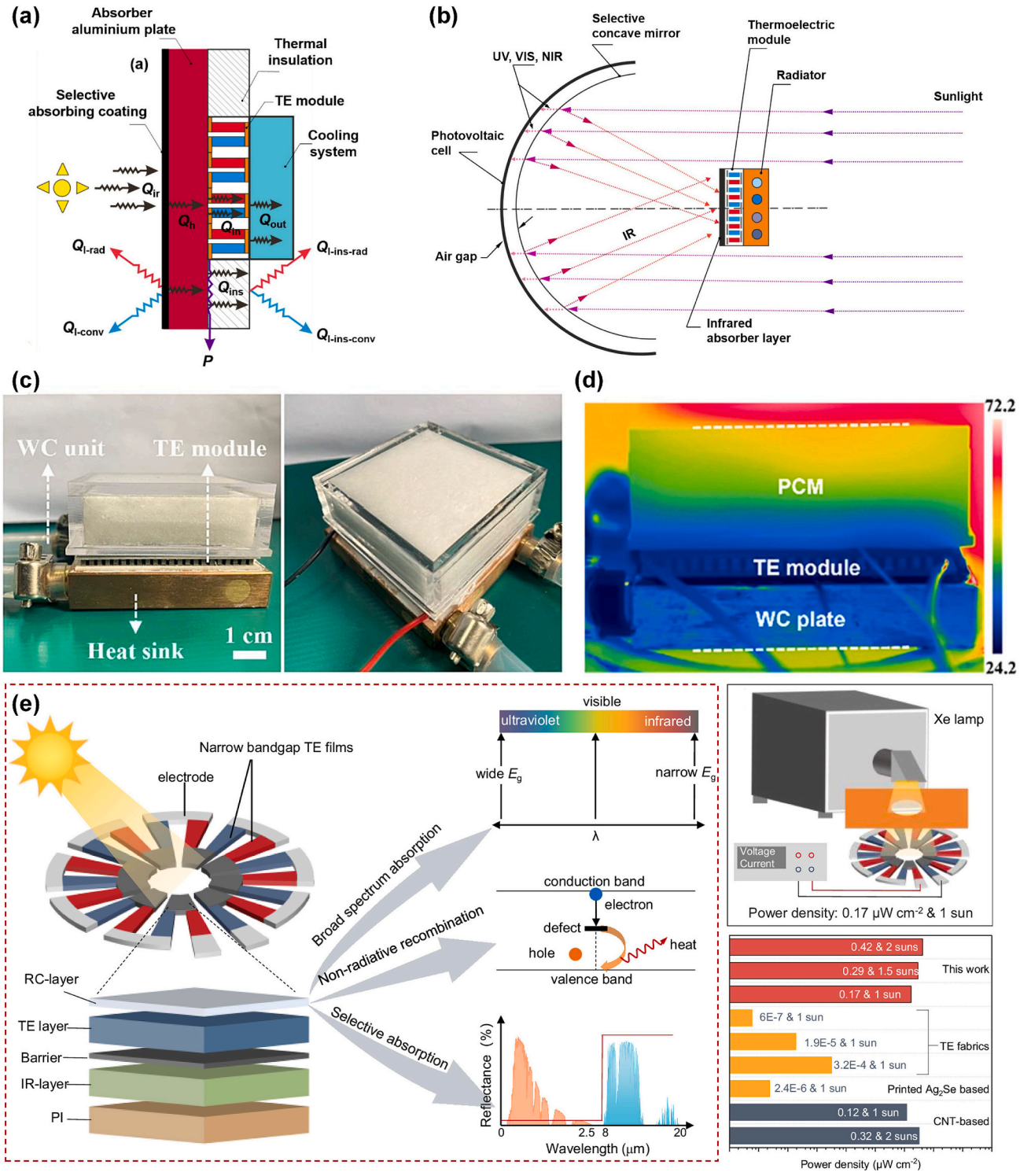

Fig. 9. Solar TEG heat collectors. (a) Heat-transfer model of an STEG. (b) Hybrid PV-STEG system. Reproduced by permission [147]. Copyright 2025 Elsevier. (c) STEG with a PCM-assisted collector. (d) Infrared temperature distribution of the PCM–STEG system. Reproduced by permission [153]. Copyright 2024 Elsevier. (e) All-in-one Ag_2Se thin-film solar thermoelectric structure. Reproduced under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [154]. Copyright 2026 The Authors.

4.2. Solar TEG Heat Collectors

Solar TEG (STEG) collectors must transform low-density, intermittent irradiation into a stable thermal boundary. Their hot-side source-access resistance is not only a conduction or convection problem, but an opto-thermal problem involving solar absorption, spectral selectivity, radiative emission, convective loss, heat spreading, and temporal storage [141]. For unconcentrated systems, the first priority is to absorb solar radiation while suppressing mid-infrared radiation and hot-side convection [142]. Cotfas et al. [143] used a metal-oxide composite absorber with high solar absorptance and low thermal emittance, while Xu et al. [144] combined a femtosecond-laser-machined selective absorber with a greenhouse enclosure to reduce convective loss. These studies show that hot-side collection in STEGs is controlled by the competition between optical gain and environmental heat loss.

Under concentrated irradiation, the limiting problem shifts from absorption alone to spatial heat-flux management [145]. Nonuniform irradiation can produce local overheating, uneven module temperatures, and reduced array-level output [146]. Maksymuk et al. [147] modeled a hybrid PV-STEG system (Fig. 9(a)–(b)) with high electrical and thermal power density, but concentrated systems remain sensitive to flux distribution. Long et al. [148] showed that the optimum heat-flux distribution is not necessarily the most uniform one; rather, there is an optimal spatial pattern for efficiency under a given module and boundary condition [149,150]. Li et al. [151] further demonstrated that collector geometry must be co-optimized with transparent covers, insulation, and leg height. These findings imply that solar collectors should be designed as coupled optical/thermal/module systems [152], not as isolated absorbers.

Temporal intermittency is another defining constraint. PCMs are increasingly integrated into STEG collectors to absorb peak irradiation and release heat under weak sunlight (Fig. 9(c)-(d)) [153,155]. Such collectors transform the hot side from a passive absorber into a dynamic absorption-storage boundary. The benefit is not necessarily a higher instantaneous peak power, but a longer effective generation period and a smoother temperature difference [156]. Therefore, PCM-assisted STEGs should be evaluated by daily energy yield, usable generation duration, start-up delay, and thermal cycling stability, rather than only by peak output.

Recent designs further reduce intermediate thermal resistance by depositing photothermal layers directly onto thermoelectric elements [157]. Ma et al. [158] coated TEGs with spinel-type $Cu_{1.5}Mn_{1.5}O_4$ and reported large improvements in voltage and power density. Hou et al. [154] developed an all-in-one Ag_2Se thin-film (Fig. 9(e)) solar thermoelectric structure, generating a measurable temperature gradient under 1 sun. In PV-TEG cascades, using the PV backsheet as a direct heat-collection interface can also reduce intermediate losses and improve system efficiency [159]. However, direct photothermal integration must be interpreted carefully because output enhancement may arise from increased optical absorption, reduced contact resistance, or both.

Overall, STEG collectors are constrained by the mismatch between diffuse, fluctuating solar irradiation and the module's need for a stable high-quality temperature difference. Future solar collectors should integrate spectral selectivity, spatial flux shaping, hot-side insulation, and thermal buffering. Their performance should be reported using optical efficiency, hot-side loss, module-level temperature difference, daily energy output, and environmental conditions.

4.3. Geothermal TEG heat collectors

Geothermal TEG collectors are governed by long heat-transfer paths and finite geological thermal replenishment. Unlike exhaust or combustion systems, where heat-transfer enhancement can often increase the accessible heat flux, geothermal systems are limited by the rate at which heat can be conducted or convected from the surrounding ground or reservoir. The collector must therefore minimize intermediate temperature loss while avoiding local thermal drawdown.

Passive phase-change collectors, including heat pipes and thermosyphons, are attractive because they can extract heat continuously without pumping power [160]. Pascual et al. [161] reported a 36% power increase using a lower-resistance water-based heat exchanger, and comparative studies indicate that heat-pipe collectors can outperform thermosyphon variants [162]. Field demonstrations, such as the Timanfaya geothermal TEG installation (Fig. 10 (a)–(c)), show the engineering feasibility of this route [163]. However, increasing the number of modules does not necessarily scale output proportionally. In the reported field system, increasing module count reduced per-module power, indicating that local heat spreading and geological replenishment, rather than module capacity alone, limit scalability. This is a clear example of bottleneck transfer from the module to the heat source.

For deeper resources, long gravity heat pipes and wellbore collectors move the heat-exchange zone underground to reduce surface-side losses [165,166]. Coaxial wellbore systems with integrated TEG sections have been modeled to produce substantial net output at depth [167], and laboratory prototypes have been used to project larger-scale performance [168]. Modular systems using hot water (Fig. 10(d)) or two-phase thermosyphons further aim to distribute limited geothermal heat more uniformly across module arrays [164,169]. These configurations reduce intermediate thermal resistance, but their viability depends on wellbore geometry, working-fluid behavior, reservoir temperature, and long-term thermal recovery.

For shallow and low-grade geothermal sources, direct-contact collectors are often more relevant than large heat exchangers. Spring water, soil, rock walls, and underground tunnels provide limited but persistent temperature differences. Flexible TEGs attached to copper collectors [170], gravity-heat-pipe soil harvesters [171], conductive rods for agricultural IoT [172], and mine-wall TEGs [173] all follow the same principle: shorten the heat path and reduce source-side contact loss. In these systems, the critical variable is not maximum heat-transfer coefficient, but the stability of the local thermal field and the durability of the source contact.

Overall, geothermal collectors should be designed around dynamic thermal-capacity matching. Static thermal-resistance minimization is insufficient if continuous extraction cools the local source faster than heat can be replenished. Future studies should report long-duration output, source-temperature recovery, thermal drawdown, module spacing, and energy yield per occupied thermal field, rather than only initial power.

4.4. Combustion-, biomass-, and stove-driven TEGs

Combustion-, biomass-, and stove-driven TEG collectors operate under mixed thermal boundaries involving flame radiation, wall conduction, and flue-gas convection [30]. Their main constraints are nonuniform heat flux, rapid thermal fluctuation, ash deposition, and local overheating. The collector must therefore not only deliver heat, but also stabilize and condition a harsh combustion boundary before it reaches the module.

One route is to convert the combustor or stove wall into the collector. In biomass stoves, wall-integrated Cu plates can provide direct module mounting and substantially improve output [174]. However, the wall thickness and thermal mass must be carefully selected. Increasing plate thickness can reduce module mounting temperature and prevent overheating, but it also increases thermal inertia and may reduce heat-transfer responsiveness [175]. Wall-integrated collectors therefore involve a trade-off among heat capture, thermal buffering, hot-side temperature control, and structural robustness.

A second route is hot-side thermal buffering. In biomass boilers, ash deposition progressively increases surface resistance and reduces output. Usón et al. [176] showed that output decay depends strongly on fuel type because ash accumulation changes the collector boundary. In small off-grid stoves, the dominant issue is often temporal fluctuation

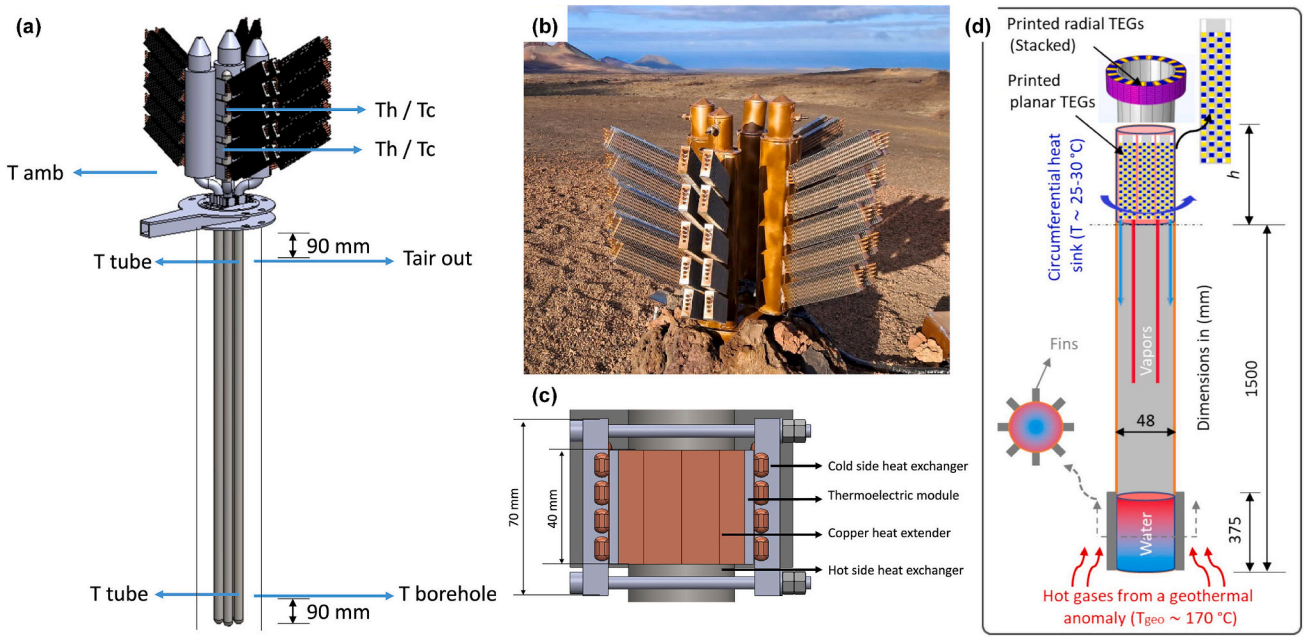

Fig. 10. Geothermal TEG heat collectors. (a) System design of the geothermal TEG. (b) Photograph of the geothermal TEG. (c) Assembly of the geothermal TEG. Reproduced by permission [163]. Copyright 2025 The Authors. (d) Geothermal TEG integrated with a two-phase thermosyphon collector. Reproduced by permission [164]. Copyright 2026 The Authors.

rather than fouling. Sensible-heat storage blocks can sustain airflow and electrical output after fuel feeding stops [177]. In both cases, the collector functions as a thermal stabilizer rather than a simple heat-transfer enhancer.

In gas-fueled and micro-combustion TEGs, the collector is increasingly integrated with the burner, heat spreader, fins, and air-preheating structure. As shown in Fig. 11(a) and (b), Li et al. [178] developed a self-powered gas-fueled combustion TEG by optimizing fin distribution and

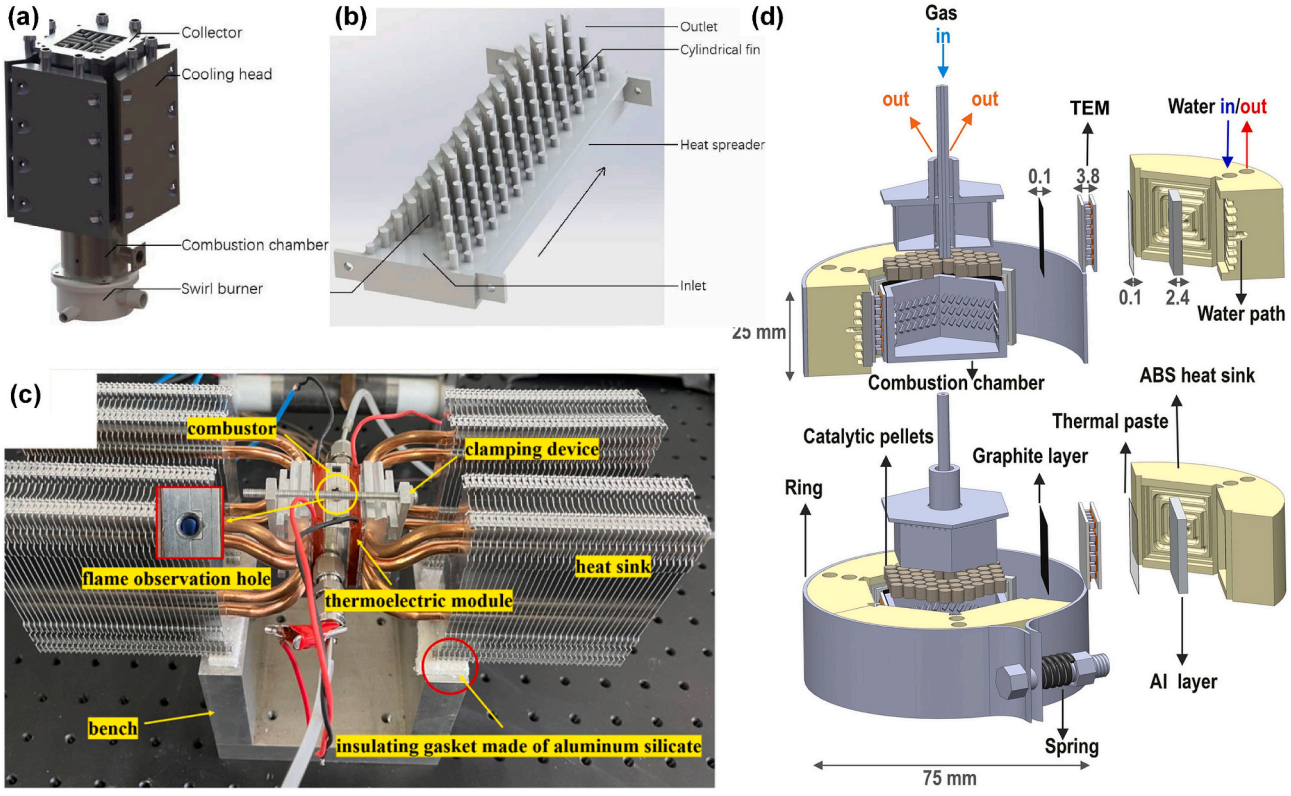

Fig. 11. Heat collectors for micro-combustion-powered TEGs. (a) Prototype of a combustion-powered TEG. (b) Design of a combustion-powered TEG collector. Reproduced by permission [180]. Copyright 2023 Elsevier. (c) Radiative heat-transfer collector for combustion-powered TEG. Reproduced by permission [181]. Copyright 2023 Elsevier. (d) Combustion-powered TEG with a hexagonal finned collector. Reproduced under a Creative Commons Attribution 4.0 International License [182]. Copyright 2023 The Authors.

heat-spreader thickness, while subsequent premixed and swirl-burner systems achieved higher compact power outputs through integrated combustion–collection design [179,180]. These systems show that, for compact combustors, thermal management cannot be separated from combustion stability and flow organization.

Non-contact radiative collection provides another strategy. As shown in Fig. 11(c) and (d), porous burners and catalytic combustion chambers can heat the TEG hot side radiatively without strong mechanical contact or severe combustion disturbance [181–183]. The advantage is a cleaner and more stable thermal boundary, especially when direct wall contact would cause local overheating or fouling. The limitation is that performance becomes sensitive to radiation distance, view factor, wall temperature, surface emissivity, and geometric alignment.

Overall, combustion-driven collectors should be evaluated by stable net output, hot-side temperature control, fouling resistance, thermal cycling durability, and combustion-system compatibility. Future designs should combine integrated combustor–collector structures, radiative coupling, and thermal buffering, but must avoid overemphasizing peak power under clean steady flames.

4.5. Industrial waste heat TEGs

Industrial waste heat sources, including hot walls, moving solids, steel wires, slag, and exhaust streams, are characterized by spatial temperature heterogeneity and strict non-interference requirements. The collector must recover heat without disrupting the primary industrial process. Therefore, industrial TEG design is primarily a problem of thermal-field matching rather than maximum area coverage [184].

Selective placement can be more effective than expanding module number. In a steel-wire production line, Miao et al. [185] showed that targeting the optimal heating section achieved better cost effectiveness than simply increasing installation footprint, even when the larger installation produced higher total output. This result highlights a general principle: module coverage should follow the spatial distribution of available exergy, not the available physical area alone [186].

For high-temperature or moving sources, local overheating and nonuniform radiation become critical. Radiation shields can improve both power output and efficiency by controlling radiative exchange [187]. Non-contact collectors for moving blast-furnace slag avoid mechanical interference while using interfacial materials to improve temperature uniformity [188]. In such systems, the collector must transform a moving, nonuniform, and often inaccessible heat source into a stable module boundary.

For medium- and low-grade industrial walls, over-installation may deplete the local thermal field. Górszczak et al. [189] found that a smaller module array could outperform a denser one because dense packing reduced the available surface temperature. Passive heat spreaders, thermosyphons, and micro heat-pipe arrays can consolidate dispersed heat and improve module temperature difference [190,191]. However, their benefit should be judged against added cost, mass, installation complexity, and maintenance requirements.

Overall, industrial TEG collectors should not be designed by maximizing contact area or module count. The key is to match collector placement and heat-spreading architecture to the actual thermal field while preserving the host process. Future work should report net output, heat-source disturbance, installation cost, surface-temperature recovery, maintenance requirements, and durability under industrial contamination or vibration.

4.6. Building-integrated TEGs

Building-integrated TEG collectors operate under extremely low temperature differences, strong diurnal and seasonal fluctuations, and strict architectural constraints. They are therefore better understood as micro-power harvesters or auxiliary energy-recovery components, rather than bulk power-generation devices [192,193]. Their collector designs usually take the form of wall cavities, thermal buffers, duct attachments, or chimney-mounted exchangers [194,195].

Envelope-integrated systems exploit indoor–outdoor temperature differences but are strongly dependent on climate, orientation, insulation state, and occupant operation. Thermoelectric wall cavities can generate modest areal output under favorable cold-region conditions [196], while PCM-assisted thermoelectric bricks can delay heat release and extend operating time [197]. However, their low efficiency and weather dependence limit their role to distributed sensing or low-power building monitoring. For building envelopes, preserving indoor comfort and envelope performance is usually more important than maximizing electrical output.

Building service systems provide higher-quality heat sources. Exhaust ducts, kitchen hoods, and chimneys offer more stable temperature differences than external walls. Counterflow exchangers with porous metal meshes [198], commercial kitchen exhaust recovery concepts [199], and direct mounting on chimney walls [200] can provide higher output because they reduce source-access resistance and use hotter waste streams. Nevertheless, their scalability is constrained by existing duct geometry, pressure-drop limits, maintenance access, and fire-safety requirements.

Overall, building-integrated collectors should be matched to the thermal hierarchy of the building. Envelope-based TEGs are most suitable for autonomous sensors and niche low-power devices, whereas service-duct and chimney-based systems offer more meaningful heat-recovery potential. Future studies should report annual energy yield, indoor-comfort impact, climate dependence, installation constraints, and net benefit relative to the building's primary thermal-management function.

5. Thermal management of heat sinks

The cold-side heat sink determines how effectively unconverted heat is rejected from the thermoelectric module to the ambient sink. In the thermal-resistance-chain framework, it mainly controls the cold-side heat-rejection resistance $R_{sink,c}$, cold-side temperature uniformity, and the realizable temperature difference across the thermoelectric legs. Although heat sinks share the same thermodynamic function, their effective boundary quality is strongly scenario-dependent. Air, liquid coolant, phase-change media, sky radiation, evaporation, and wearable microclimates differ substantially in heat capacity, heat-transfer coefficient, available area, parasitic cost, and environmental stability.

Therefore, the design target is not simply to minimize the cold-side temperature. A heat sink that lowers the cold-side temperature by consuming excessive fan or pump power may reduce net system output. Similarly, a passive sink with zero auxiliary power may be ineffective if its cooling capacity is too small or too sensitive to weather. Cold-side thermal management should therefore be evaluated through four coupled criteria: heat-rejection resistance, temperature uniformity, parasitic energy consumption, and dynamic or environmental stability. Based on these criteria, this section reviews air-cooled, liquid-cooled, two-phase, PCM-assisted, radiative/evaporative, and wearable-integrated cold ends.

5.1. Air-cooled heat sinks for TEGs

Air cooling is attractive because of its structural simplicity, low cost, and compatibility with compact TEG systems. Its main limitation is the low convective heat-transfer coefficient of air. Natural convection usually provides limited heat rejection, whereas forced convection improves cooling at the expense of fan power. As a result, the relevant objective is net electrical output rather than cold-side temperature reduction alone. Pujol et al. [201] and Chen et al. [202] showed that net power exhibits an optimum with airflow rate; beyond this point, the additional fan power outweighs the thermoelectric gain.

For conventional finned heat sinks, the key issue is how to disturb the

thermal boundary layer without imposing excessive aerodynamic resistance. Gürcan et al. [203] compared straight, slotted, and perforated fins with aluminum foams, as shown in Fig. 12(a). Slotted fins improved output moderately, while perforated aluminum foam produced a larger power increase but introduced a stronger pressure-drop penalty. This illustrates a general constraint of air-cooled TEGs: local heat-transfer intensification is useful only when the additional flow resistance remains small.

Recent work has therefore moved from empirical fin modification toward topology-optimized flow routing. Park et al. [204] developed an additive-manufactured, topology-optimized forced-convection heat sink that reduced thermal resistance under a fixed pressure-drop constraint, as shown in Fig. 12(b). For fanless systems, Xie et al. [205,207] optimized fin geometry according to buoyancy-driven flow behavior, improving power output relative to straight fins, as shown in Fig. 12(c). System-level aerodynamic control, such as exhaust-induced self-cooling or air deflectors that suppress thermal backflow, further expands air cooling beyond the heat sink itself [206,208]. Future air-cooled designs should report fan power, pressure drop, net output, and cold-side temperature uniformity; otherwise, apparent heat-transfer enhancement may not correspond to system-level gain.

5.2. Liquid-cooled heat sinks for TEGs

Liquid cooling provides higher heat-transfer coefficients and greater thermal capacity than air cooling, making it suitable for high-heat-flux and compact TEG systems. Its main penalty is hydraulic power consumption, together with added complexity in sealing, pumping, corrosion control, and maintenance. As with forced air cooling, increasing coolant flow rate produces diminishing returns. Once the cold-side thermal resistance is sufficiently reduced, further flow acceleration mainly increases pumping loss. Therefore, liquid-cooled TEGs should be optimized by net power rather than coolant flow rate or cold-side temperature alone [209].

Flow organization is often more effective than brute-force flow acceleration. In annular automotive TEGs, optimizing coolant flow rate increased maximum net power substantially [210], while indirect-contact and counterflow configurations improved cold-side boundary matching [211]. Under restricted volume, microchannels shorten the heat-transfer path and enlarge the specific heat-transfer area. Compared with macro-channel designs, microchannel cold plates can improve net power while reducing coolant demand [212]. Internal metastructures and local flow disturbers further thin the thermal boundary layer, as shown in Fig. 13(a), but their benefits must be weighed against pressure drop and manufacturing complexity [213].

For micro-TEGs, manifold microchannels provide an especially strong cold boundary because they distribute coolant over very small active areas. As shown in Fig. 13(b), manifold microchannel heat sinks have been applied to Bi_2Te_3, Mg_3Sb_2, and SnSe micro-devices, producing large improvements over uncooled baselines [214]. However, extreme miniaturization can create severe hydraulic resistance, clogging risks, and fabrication constraints. Future liquid-cooled heat sinks should therefore report pumping power, flow rate, pressure drop, coolant temperature rise, thermal uniformity, and net electrical output. Without these data, high cooling performance cannot be fairly translated into TEG system performance.

5.3. Two-phase heat spreaders and heat sinks

Two-phase cold-side devices, including heat pipes, vapor chambers, and micro heat-pipe arrays, should not be viewed simply as stronger heat sinks. Their main value is to reduce cold-side spreading resistance,

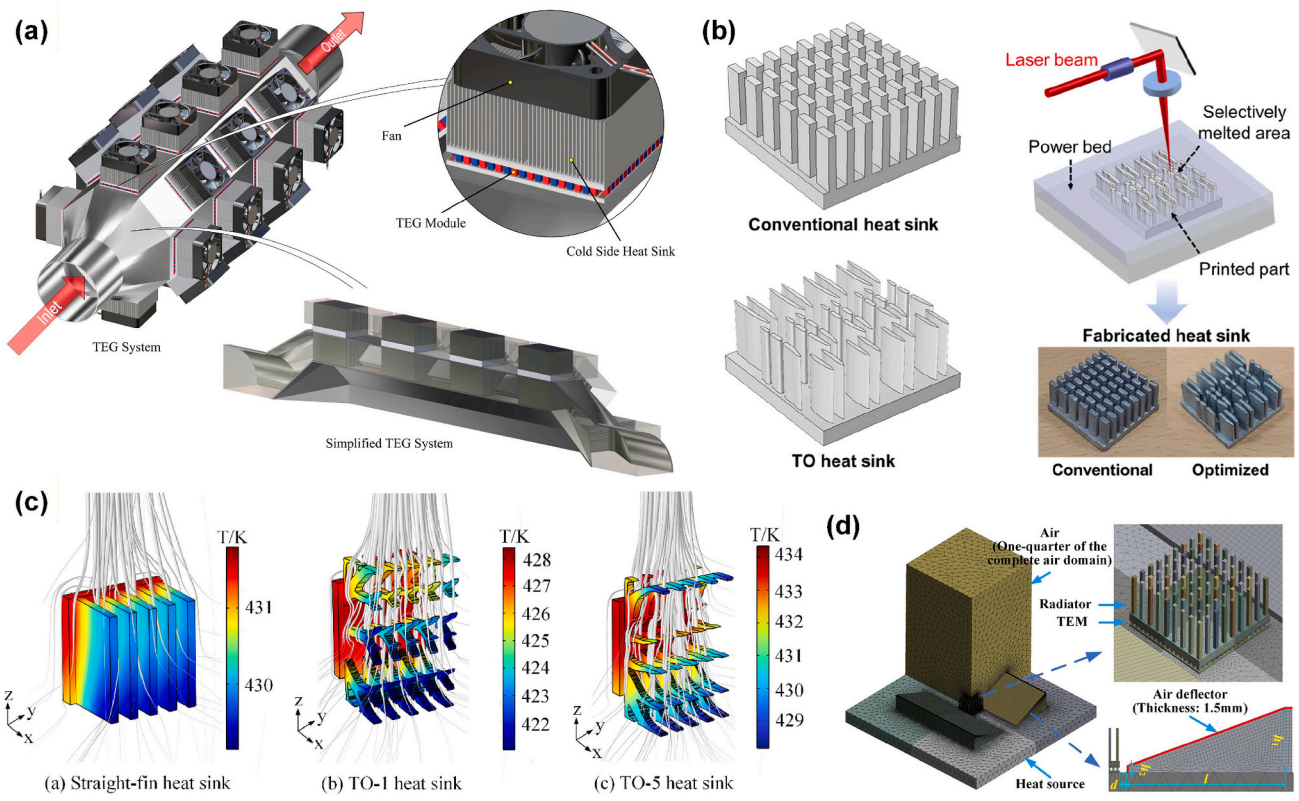

Fig. 12. Progress in air-cooled TEG heat sinks. (a) Fin-and-fan forced-convection TEG system. Reproduced by permission [203]. Copyright 2025 The authors. (b) Additive-manufactured topology-optimized heat sinks. Reproduced by permission [204]. Copyright 2025 Elsevier. (c) Topology-optimized fins based on the Grashof number. Reproduced by permission [205]. Copyright 2025 Elsevier. (d) Hybrid TEG heat sink combining fins and an air deflector. Reproduced by permission [206]. Copyright 2026 Elsevier.

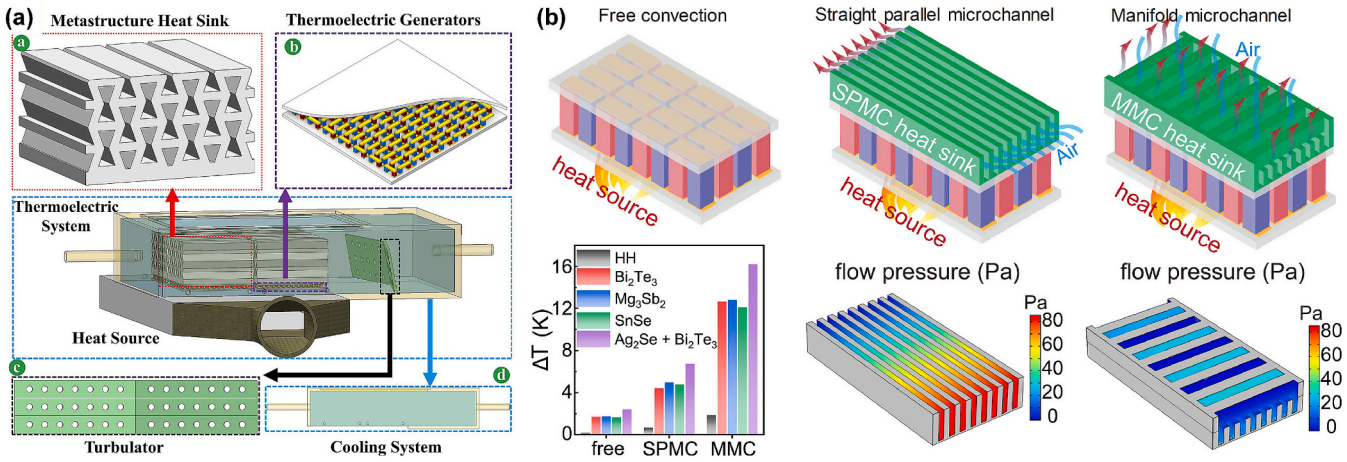

Fig. 13. Progress in water-cooled TEG heat sinks. (a) Water-cooled TEG heat sink with a flow-disturbing element. Reproduced by permission [213]. Copyright 2025 The Authors. (b) Manifold microchannel heat sink for micro-TEGs. Reproduced by permission [214]. Copyright 2024 Elsevier.

improve isothermality across the module footprint, and decouple the cold junction from the final ambient rejection site [215,216]. This is particularly useful when the TEG is exposed to localized hot spots, uneven module heating, or spatially constrained cooling surfaces.

Conventional heat pipes transport heat from the module cold side to a remote condenser with low effective thermal resistance [217]. They also provide transient buffering under fluctuating thermal loads [218]. However, their continuous performance is ultimately limited by the condenser-side heat rejection, which often still requires air cooling or another external sink. Thus, a heat pipe can reduce the internal cold-side spreading loss but cannot eliminate the need for final heat rejection.

Micro heat-pipe arrays are used to reduce the geometric mismatch between cylindrical heat pipes and flat TEG modules. Their performance depends strongly on bend angle, length, capillary return, gravity orientation, and working-fluid distribution [219]. Optimized U-shaped micro heat-pipe arrays have produced notable performance gains [220], but these geometries cannot be generalized without considering orientation sensitivity and capillary limits. Future work should distinguish evaporator-side improvement from condenser-side limitation and should report start-up behavior, allowable heat load, orientation dependence, and long-term sealing reliability.

5.4. PCM-assisted cold-side buffering

Phase-change materials provide latent heat storage that delays cold-side temperature rise during pulsed or intermittent heat input [221]. Unlike air or liquid cooling, a PCM heat sink does not provide unlimited steady heat rejection. Its function is transient buffering: it stores heat for a finite period and must later release that heat during regeneration. PCM-assisted cold ends are therefore most suitable for short-duration operation, periodic thermal loads, or systems where suppressing temperature fluctuation is more valuable than maximizing steady-state heat rejection.

The main limitation of pure PCMs is low thermal conductivity [222]. Conductive matrices, metal foams, and fins are commonly introduced to accelerate heat diffusion and improve latent heat utilization [223,224]. Copper-foam-enhanced PCMs can reduce cold-side temperature and improve power output relative to simple fan-assisted designs [225]. The matrix material, porosity, pore density, and fin configuration strongly influence melting uniformity and energy output [226–228]. However, higher matrix content can reduce latent heat capacity, increase mass, and complicate regeneration.

PCM heat sinks should therefore not be judged by peak steady-state power. More appropriate metrics include cycle-averaged energy, duration above usable voltage, cold-side temperature fluctuation, melting/solidification completeness, regeneration time, and cycling durability [229]. From the thermal-resistance-chain perspective, PCM buffering is valuable when it improves time-averaged temperature-difference utilization without creating excessive mass or thermal inertia.

5.5. Radiative-cooling heat sinks for TEGs

Radiative cooling offers passive heat rejection without fan or pump power. Its cooling capacity, however, is limited by atmospheric radiation exchange and is highly sensitive to humidity, cloud cover, solar absorption, sky view factor, and parasitic environmental heat gain [230]. Therefore, maximizing infrared emissivity alone is insufficient. The emitter must also suppress solar absorption, maintain good thermal contact with the module, and provide enough area to match the heat load.

Recent work has shifted from bulky radiative cavities to thin porous or polymeric emitters that can be integrated closer to the TEG cold side [231]. Zhang et al. [232] demonstrated a porous P(VdF-HFP) copolymer flat emitter for building-related cooling, while Hei et al. [233] applied an advanced radiative film directly onto the cold side to increase TEG power density. Adaptive emitters, such as PDMS films containing hollow silica microspheres and thermochromic pigments, further adjust radiative behavior with temperature, as shown in Fig. 14(a) [234]. These designs are promising for low-heat-flux systems, but their effectiveness depends strongly on local climate.

Because pure radiative cooling has finite capacity, hybrid environmental sinks are increasingly used. Tandem radiative, evaporative, and PCM cooling structures combine long-wave emission, water evaporation, and latent heat storage, as shown in Fig. 14(b) [235]. Free-surface evaporation and hydrogel-assisted evaporation can further enhance passive cooling [236,237]. Nevertheless, evaporative systems introduce dependence on humidity, water retention, drying rate, and maintenance. Future studies should report meteorological conditions, solar exposure, humidity, sky view factor, cooling area, water consumption if applicable, and daily energy yield. Without these conditions, radiative or evaporative cooling results are difficult to compare across locations.

5.6. Special heat sinks for wearable TEGs

Wearable TEGs impose the strictest cold-side integration constraints. The heat sink must be thin, lightweight, flexible, breathable, and comfortable, while operating under low airflow and variable skin contact. In these systems, the cold end is rarely an external component. It is usually integrated into the substrate, electrode, textile, coating, or encapsulation layer [238].

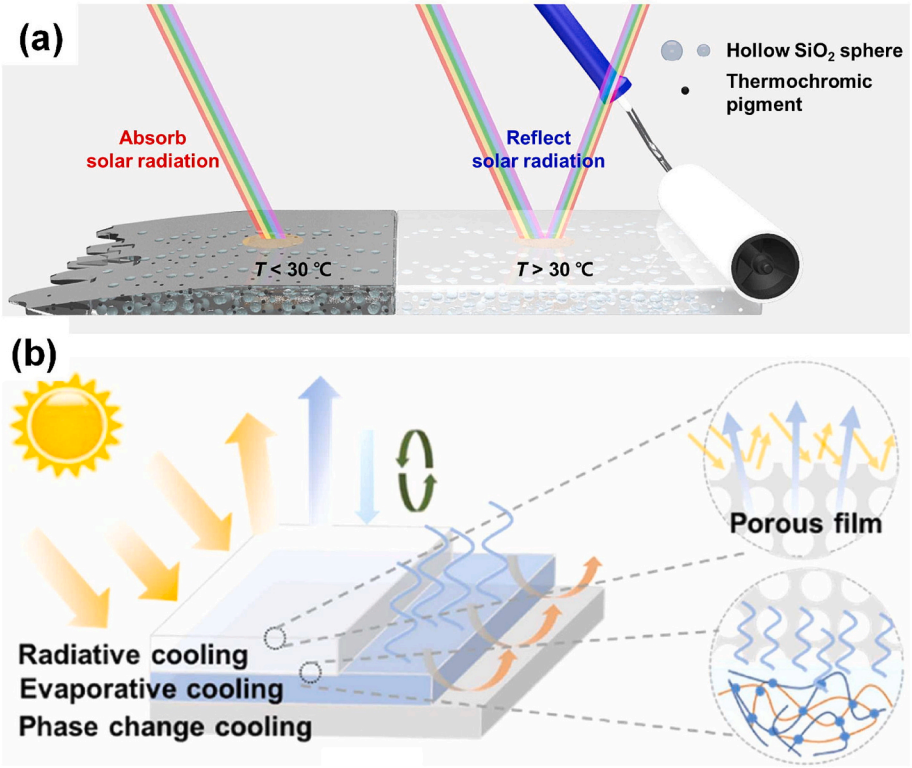

Fig. 14. Radiative and evaporative heat sinks. (a) Adaptive radiative-cooling heat sink. Reproduced by permission [234]. Copyright 2023 Elsevier. (b) Tandem radiative/evaporative/PCM cooling structure. Reproduced by permission [235]. Copyright 2024 Elsevier.

Porous metallic skeletons can serve simultaneously as electrodes, structural supports, and convective heat spreaders. Lv et al. [239] used a porous sandwich substrate combined with a copper-foam sink to improve resting and airflow-assisted power density. Zhao et al. [82] integrated the cold end and top electrode into a copper foam layer, shortening the heat path. For ultrathin flexible modules, radiative coatings can replace bulky heat sinks. Gobpant et al. [240] applied a radiative cooling layer to a flexible TEG, while Li et al. [241] directly deposited a radiative coating onto the cold surface, reducing adhesive-related resistance and delamination risk.

Hydrogel-based cold ends introduce an evaporative route analogous to perspiration. Wang et al. [242] combined a copper foam skeleton with a porous hydrogel to improve natural- and forced-convection performance. However, hydrogel cooling depends on ambient humidity, water retention, and long-term dehydration. More broadly, wearable heat sink studies remain difficult to compare because wearing position, skin temperature, contact pressure, bending state, airflow, sweat, and ambient humidity vary substantially. Future evaluation should include total thickness, areal mass, air permeability, contact pressure, bending durability, sustained power under natural convection, and long-duration on-body testing.

6. Key challenges and future directions in TEG thermal management

Interfaces and packaging, hot-side heat collectors, and cold-side heat sinks act on different segments of the thermal-resistance network, but they all determine the same system-level quantity: the effective temperature difference established across the thermoelectric legs. The studies summarized in Table 3 show that most reported advances still optimize a single segment of this chain, such as a heat exchanger, a TIM, a cold plate, or a thermal buffer, while the remaining segments are treated as fixed boundary conditions. This component-centered strategy has produced many local improvements, but it also makes the reported gains difficult to compare and often insufficient to predict system-level performance. The central challenge in TEG thermal management is therefore to move from isolated resistance reduction to full-chain resistance diagnosis, allocation, and coordination. (See Fig. 15)

6.1. Dominant-resistance diagnosis

The first challenge is the lack of a unified method for identifying the dominant thermal bottleneck. In the thermal-resistance framework established in Section 2, the nominal temperature difference is divided among hot-side source-access resistance, interface/package resistance, the thermoelectric branch, cold-side heat-rejection resistance, and parasitic heat leakage. The effective working temperature difference is only the remaining portion imposed on the thermoelectric legs. Therefore, thermal management should begin by determining where the temperature difference is actually lost.

The dominant resistance varies strongly with application scenario. Automotive exhaust and industrial fluid systems are often constrained by gas- or liquid-side convection together with pressure-drop penalties. Solid waste-heat and wall-coupled systems are more sensitive to contact and spreading resistances. Solar TEGs are governed by optical access, hot-side loss, flux nonuniformity, and environmental cold-sink capacity. Geothermal TEGs may be limited by geological heat replenishment rather than module-side heat transfer. Wearable systems are dominated by body-side conduction, nonconformal contact, and lateral heat leakage before conventional heat sink design becomes relevant.

Future studies should therefore report a temperature-loss budget rather than only output improvement. A useful thermal-management strategy should clarify which resistance is dominant, how much of the available temperature difference it consumes, and whether reducing it shifts the bottleneck to another part of the chain. Without this diagnosis, heat-transfer enhancement may be applied to a non-limiting segment and produce little system-level benefit.

Table 3
Summary of representative heat collector and heat sink designs across multi-scenario TEG applications.

Application	Method	Key features	Heat collector	Heat sink	$T_{\infty,h}/T_{\infty,c}$	T_h/T_c	Performance	Ref.
Automotive exhaust	Sim.	Multi-objective optimization of square pin fins to minimize thermal stress while maintaining output	Square pin-fin collector	Water-cooled sink	498.15 K exhaust/ 303.15 K water	ΔT_{ave} = 132.54	Hot-side heat transfer increased by 8.57%; max thermal stress reduced by 11.78%.	[243]
Automotive exhaust	Sim.	Evaluates bi-planar, C-shaped, and U-shaped surface dimple topologies for heat exchangers.	Dimpled exchanger with baffles	Water-cooled sink	808.15 K exhaust/ 363.75 K water	512 K/374 K	$P_{net,max}$ = 463.02 W; $\eta_{net,max}$ = 3.91%	[244]
Automotive exhaust	Sim.	Integrates variable conductance heat pipes (VCHPs) to passively spread excess heat to under-heated regions	Corrugated stainless-steel pipe with two-row VCHPs	Water-cooled sink	873.15 K exhaust / 296.15 K water	–	P_{max} = 310 W, average fuel saving of 4%.	[245]
Automotive exhaust	Sim. + Exp.	Investigates dynamic response and thermal hysteresis under CLTC using a polygonal exchanger	Octagonal exchanger with sickle-shaped fins	Water-cooled sink	809 K exhaust / 369 K water	439.85 K/ 377 K	P_{ave} = 24.48 W and η = 1.12% over the entire CLTC	[246]
Automotive exhaust	Sim.	Streamwise gradient distribution of dimpled fins to match exhaust temperature decay and boost net power	Dimpled-fin exchanger	Water-cooled sink	546.73 K exhaust / 300 K water	T_h = 430 K	$P_{net,max}$ = 62.46 W; η, P, V, and P_{net} increased by 16.07%, 28.95%, 13.56%, and 10.09%	[247]
Automotive exhaust	Sim. + Exp.	Global optimization of an octagonal heat exchanger with sickle-shaped fins; introduces vehicle-compatibility metrics	Octagonal exchanger with sickle-shaped fins	Water-cooled sink	600 K exhaust / 300 K water	420.16 K/ 318.7 K	P_{net} and η_{net} increased by 6.75% and 7.06%	[248]
Automotive exhaust	Exp.	Experimental, exergy, and economic evaluation of an automotive TEG	Rectangular gas exchanger	Water-cooled sink	673 K exhaust / 297 K water	–	$P_{net,max}$ = 12.58 W; $\eta_{net,max}$ = 1.76%; exergy efficiency of 3.17%.	[249]
Automotive exhaust	Sim.	Introduces split fins to conventional plate fins for enhanced heat transfer and temperature uniformity	Split-fin collector	Water-cooled sink	550 K exhaust / 300 K water	T_h = 441.73 K	P, V, η, and V-uniformity increased by 23.71%, 11.28%, 11.29%, and 2.23%.	[124]
Automotive exhaust	Sim.	Proposes a diverging flow-splitting pillar with an increasing cross-sectional area along the flow direction	Hexagonal exchanger with a central splitting pillar	Water-cooled sink	750 K exhaust / 300 K water	T_h = 474.14 K	P and η increased by 9.38% and 5.81%; V-uniformity of 0.975	[250]
Automotive exhaust	Sim.	Couples TEG with a diesel particulate filter for exhaust energy recovery	Three-layer porous-media exchanger	Air-cooled finned sink	700 K exhaust / ambient air	576 K/403 K	P_{max} = 8.49 W	[251]
Motorcycle exhaust	Sim.	Adds a converging duct to the cold side to optimize airflow distribution for enhanced forced-air cooling	Guide-fin and perforated-plate collector	Converging-duct air-cooled sink	712 K exhaust / 303 K ambient air	T_h = 341 K; ΔT = 51.03 K	P_{max} = 16.36 W; T-uniformity increased by 13.78%.	[252]
Solid oxide fuel cell waste heat	Sim.	Optimizes a pin-fin exchanger specifically for the high-temperature, low-flow-rate exhaust of Solid oxide fuel cells	Pin-fin heat exchanger	Water-cooled sink	1073 K exhaust / 300 K water	ΔT = 400 K	P_{net} = 150.3 W; η = 5.06%; P_{net} and η_{net} increased by 400.3% and 142.16%	[253]
Hybrid Photovoltaic -TEG	Sim.	Simultaneously regulates the Photovoltaic -TEG temperature field using heat pipes and radiative cooling	Heat-pipe	Radiative cooler	1000 W m^{-2} irradiance/ ambient air	–	Energy yield increased by up to 7.2%; LCOE of 0.065–0.089 USD kWh^{-1}	[254]
Hybrid Photovoltaic -TEG	Sim.	Multi-objective optimization of a building-integrated PV-TEG coupled with microencapsulated PCM	PV panel collector	Finned microencapsulated PCM sink	1000 W m^{-2} irradiance / 298 K ambient air	ΔT_{max} = 6.54 K	Optimal phase-change temperature maintained at 293–313 K	[255]
Hybrid Photovoltaic -TEG	Exp.	Novel hybrid photovoltaic/ thermal collector integrating a TEG with an active water-spray cooling system	Black dye and graphene nanoparticle absorber	Active water spray + fan-assisted finned sink	995.7 W m^{-2} irradiance / 313 K ambient air	–	P_{max} = 5.6 W	[256]
Solar	Sim.	Compares hot-side vs. cold-side PCM configurations and performs topology optimization on the PCM structure	PCM-impregnated structure	PCM sink	900 W m^{-2} irradiance / 298 K ambient air	ΔT = 2.34 K	P increased by 23.5%	[257]

(*continued on next page*)

Table 3 (*continued*)

Application	Method	Key features	Heat collector	Heat sink	$T_{\infty,h}/T_{\infty,c}$	T_h/T_c	Performance	Ref.
Solar	Sim. + Exp.	Synergizes a Fresnel lens, solar tube, heat pipe, and tracking system for enhanced thermal control	Fresnel lens + solar tube + heat pipe	Nanofluid (water, Al_2O_3, TiO_2)	920 W m^{-2} irradiance / coolant	400 K/ 356 K	V_{max} = 2.76 V; η = 1.06%; P_{max} = 1.058 W	[258]
Solar	Sim. + Exp.	Collects heat via a solar glass tube and heat pipe, integrated with an active cooling system	Solar tube + heat pipe + thermal cavities	Water-cooled sink	662 W m^{-2} irradiance / 298 K water	351.8 K/ 318.4	V_{max} = 7.52 V; P_{max} = 5.236 W	[259]
Solar	Sim.	Uses graphene nanofluid to cool a segmented variable-cross-section TEG; evaluates exergy and thermal stress	Solar concentrating collector	Nanofluid (graphene, Al_2O_3, or Fe_3O_4)	595 W m^{-2} irradiance / 2 m s^{-1} coolant	–	P = 221.77 mW; exergy efficiency of 8.99%; max thermal stress of 1.35 GPa	[260]
Solar	Exp.	Utilizes edible oil as a PCM for thermal storage, enhanced with Fe_2O_3 nanoparticles or carbonized sawdust	Bio-based PCM (coconut oil, tallow) optical absorber	Finned sink	900 W m^{-2} irradiance / 298 K ambient air	308.8 K/ 304.5 K	Thermal storage efficiency increased by up to 62%	[261]
Solar	Sim. + Exp.	Vertically oriented all-weather TEG utilizing double-sided reflectors	Absorbing coating + silvered-glass reflector	Radiative cooling + Infrared reflection	Solar irradiance / ambient air	ΔT =5.6 K	$P_{ds,max}$ = 363.42 mW m^{-2}	[262]
Solar	Sim.	Angle-independent evacuated U-tube collector coupled with a TEG featuring spatially non-uniform thermal conductivity	Evacuated U-tube solar collector	Natural-convection sink	1000 W m^{-2} irradiance / 298 K ambient air	ΔT_{max} = 140 K	P_{ds} = 26.1 W m^{-2}, η = 2.61%	[263]
Solar	Sim.	Couples radiative and evaporative cooling at the cold side to enhance output across various climates	Reflector and Fresnel lens collector	Hybrid radiative-evaporative sink	Solar irradiance / ambient air	ΔT_{ave} = 84.25 K	Annual average output of 5566.76 W m^{-2}; 13.6% higher than conventional sinks	[264]
Solar	Sim.	Uses a continuous-fluid variable-temperature sink, revealing that cold-side convective resistance dominates	Solar concentrator and absorber	Water-cooled sink	471 K concentrator / 293 K water	466.5 K/ 309 K	P_{max} = 596.22 W; η_{max} = 4.98%; payback period of ~3.55 years	[20]
Solar	Sim.	Evaluates the synergistic effects of three fin types with air, water, or MWCNT-SiC nanofluid cooling	Solar soncentrating collector	Plate-fin or pin-fin sink + water/MWCNT-SiC nanofluid	1000 W m^{-2} irradiance / 293 K coolant	–	P increased by 2.6%; P_{max} = 0.35 W	[265]
Solar	Sim.	Novel hybrid strategy integrating a spectral beam splitter and confined jet impingement with microchannels	Spectral beam splitter concentrator	Finned water-cooled sink	1000 W m^{-2} irradiance / 298 K water	ΔT_{max} = 42.69 K	P_{max} = 4.13 W	[266]
Solar	Exp.	Couples a selective absorber with a superhydrophilic wiper for evaporative cooling, co-generating electricity and freshwater	Wavelength-selective absorber	Superhydrophilic cleanroom-wiper evaporative cooler	374 K absorber surface / seawater	ΔT_{max} = 14.9 K	V = 258.8 mV; P_{ds} = 0.504 W m^{-2}	[267]
Solar	Sim.	Integrates a dual-source heat pump, flat-plate solar collector, and a PCM storage tank with a wall-mounted TEG	Paraffin PCM collector	Finned sink	331 K/288 K ambient air	ΔT = 20–30 K	P_{max} = 45 W; payback period of 4.9 years	[268]
Solar	Exp.	Proposes an ultrathin, self-pumping passive wet-fabric evaporative cooling system	Parabolic dish concentrator	Ultrathin self-pumping fabric evaporative cooler	865 W m^{-2} irradiance / 296.5 K ambient air	Average 92 K/68 K; ΔT_{max} = 52.8 K	I_{max} = 5.7 mA (147% increase)	[269]
Geothermal	Exp.	Novel road-based TEG with active water circulation for summer cooling and rapid winter snow-melting	Solid-state conduction via asphalt mixture	Water-cooled sink	Asphalt pavement / 298 K water	–	P_{max} = 162.33 mW; P_{ds} = 5.136 mW cm^{-3}	[270]
Geothermal	Exp.	Multi-criteria decision-making evaluation of an ultra-long gravity heat-pipe TEG system	Ultra-long gravity heat-pipe collector	Fan-assisted forced-convection air cooler	623 K ceramic heater / 293.15 K ambient air	–	P_{max} = 28.725 W; η_{max} = 11.3%	[271]
Geothermal	Exp.	First successful deployment and continuous operation of a passive geothermal TEG in the harsh Antarctic environment	Phase-change heat-pipe collector	Phase-change heat-pipe sink	364 K shallow geothermal / 272.3 K ambient air	ΔT_{max} = 49.1 K	P_{net} = 4.36 W; annual output of 38 kWh; $(T_h—T_c)/(T_{\infty,h}$-$T_{\infty,c})$ = 0.58	[272]
Geothermal	Sim.	System-level multi-objective optimization reveals that exchanger thermal resistance bottlenecks performance before module geometry	Evaporation-condensation working fluid collector	Water/methanol heat-pipe exchanger	373 K volcanic geothermal / 253 K cold ambient	–	P increased by 3.7%–12.6%; total exchanger thermal resistance reduced by 26%	[21]
Seawater	Sim. + Exp.	PCM volume-change driven model for marine thermal energy conversion	PCM collector	Cold seawater	298 K warm water / 277 K cold water	–	Max energy output of 120 J	[273]

(*continued on next page*)

Table 3 (*continued*)

Application	Method	Key features	Heat collector	Heat sink	$T_{\infty,h}/T_{\infty,c}$	T_h/T_c	Performance	Ref.
Industrial waste heat	Sim. + Exp.	Annular structure with longitudinal vortex generator fins for secondary flow enhancement and stability	Annular collector with longitudinal vortex generator fins	Natural convection	623.15 K waste gas / 293.15 K air	$T_h = 528$ K	$P_{net} = 4.54$ W (75.81% increase); $\eta_{max} = 3.18\%$	[274]
Industrial waste heat	Sim. + Exp.	Suspends TEG above blast furnace slag; evaluates TEM height effects on flow and power generation	Finned exchanger with insulation	Water-cooled sink	1723 K blast furnace slag / 293 K water	$\Delta T_{ave} = 198.33$ K	$P_{max} = 11.93$ W	[275]
Industrial waste heat	Sim. + Exp.	Weighting analysis of multi-parameters for non-contact TEGs; fin count identified as dominant factor	Non-contact finned collector	Water-cooled sink	1073 K radiation / 293 K water	$T_h = 395.5$ K	$P_{net,max} = 65.61$ W (53.85% increase)	[276]
Low-temp water	Sim.	Plate-fin enhancement for low-grade heat recovery; evaluates fluid combinations and fin counts	Plate-fin collector	Plate-fin liquid-cooled sink	353.15 K warm water / 303.15 K coolant	$\Delta T = 22.25$ K	$P_{max} = 0.411$ W and $\eta_{max} = 0.95\%$; increased by 105.5% and 43.94%	[277]
Low-temp water	Sim.	Taguchi-optimized plate heat exchanger for flow uniformity and low pressure drop	Finned plate exchanger	Finned plate exchanger	363 K warm water / 293 K coolant	–	Single module $P = 1.2$ W, $P_{ds,net} = 51.4$ kW m^{-3}, and $\eta = 1.92\%$	[278]
Liquefied natural gas (LNG) cold energy	Sim. + Exp.	Non-frosting cryogenic TEG utilizing natural gas vapor at the hot side to prevent icing	Microchannel exchanger	Two-phase LNG pipe	298.15 K air / 120 K liquid LNG	$\Delta T_{max} = 140$ K	$P_{max} = 593.94$ W; $\eta = 1.36\%$	[279]
Liquefied natural gas cold energy	Sim.	Cryogenic LNG cold energy recovery optimization considering frosting conditions	Finned exchanger	Two-phase LNG pipe	298 K air / 112 K liquid LNG	$\Delta T_{max} = 35.6$ K	$P_{max} = 114.3$ W; $\eta = 1.05\%$	[280]
Building waste heat	Sim. + Exp.	Self-powered system integrating TEG with bio-based phase-change material	Aluminum solar collector	BioPCM sink	800 W m^{-2} irradiance/ 296.5 K ambient air	$\Delta T_{ave} = 15$ K	$P_{max} = 340$ mW	[281]
Ambient heat	Exp.	Radiative coating on a phase-change heat pipe to reduce cold-side thermal resistance under natural convection	Heat-pipe collector	Heat pipe + radiative coating	Heating plate /natural convection	–	P increased by 8.3%	[282]
Ambient heat	Sim.	Optimizes a hybrid generation system integrating concentrated solar heating and radiative cooling films	Solar-absorbing aluminum plate	Radiative-cooling aluminum plate	1000 W m^{-2} irradiance/ 296.5 K ambient air	$\Delta T_{ave} = 14$ K	$V = 386.12$ mV	[283]
Ambient heat	Exp.	Employs a high-performance radiative cooling film for day-and-night cold-side heat rejection	Soft magnetic layer	Radiative cooling film	353 K heating stage / 296.2 K ambient air	$\Delta T_{max} = 7.5$ K	$P_{ds,max} = 2.76$ W m^{-2}	[233]
Micro-combustion	Sim.	Design strategies for wall temperature uniformity and heat recovery efficiency in micro-combustors	Cross-shaped or I-shaped combustor wall	Natural-convection sink	H_2-air premixed combustion / 300 K ambient air	–	Wall temperature difference of 51.8 K, average wall temperature increased by 12 K	[284]
Micro-combustion	Sim.	Evaluates the effects of conventional vs. colorless distributed combustion on TEGs in a swirl combustor	Swirl-combustor wall	Forced-air finned sink	1400 K combustor wall / 300 K air ambient air	–	$P_{max} = 170$ W	[285]
Micro-combustion	Sim. + Exp.	H_2/NH_3 blended fuel micro-combustor integrating internal ribs and blocks	Combustor wall with internal strengthening ribs	Forced-air finned sink	H_2/NH_3 combustion / 298.15 K ambient air	–	$P_{max} = 1.065$ W (11.75% increase)	[286]
Micro-combustion	Exp.	Standalone micro-swirl combustion-powered TEG with an integrated power management system	Swirl-combustor wall	Forced-air finned sink	Butane combustion / 293 K ambient air	521.5 K/ 338 K	$P_{net,max} = 43.1$ W; $\eta_{net,max} = 3.1\%$; energy density of 393.2 Wh kg^{-1}	[287]
Electronic waste heat	Exp.	Innovative system utilizing heat pipes for both long-distance hot-side heat collection and cold-side heat rejection	Flat copper heat-pipe collector	Heat pipe + finned sink	42 W Electric heater / 1.2 m s^{-1} forced-air cooling	$T_c = 330$ K	$P_{max} = 0.8$ W; $\eta_{max} = 2\%$	[217]
Electronic waste heat	Sim. + Exp.	Novel flexible TEG integrating a sandwich substrate and directly soldered copper foam, conformally attached to high-voltage switchgears	Flexible sandwich substrate	Cubic copper-foam finned sink	353 K heat source / ambient air	–	$P_{max} = 27.487$ mW	[288]
Electronic waste heat	Sim. + Exp.	Micro-TEG equipped with a blower-assisted swirl combustor and an inverted T-shaped collector	Inverted T-shaped collector	Forced-air finned sink	585 K combustor outer wall / 298 K air	524 K/ 338 K	$P_{max} = 53.1$ W; $\eta_{max} = 3.85\%$	[289]

Note: P, V, η, T, and ΔT denote the reported power, open-circuit voltage, efficiency, temperature, and temperature difference, respectively. The subscripts "net", "ave", "max", and "ds" represent the net value, average value, maximum value, and power density, respectively.

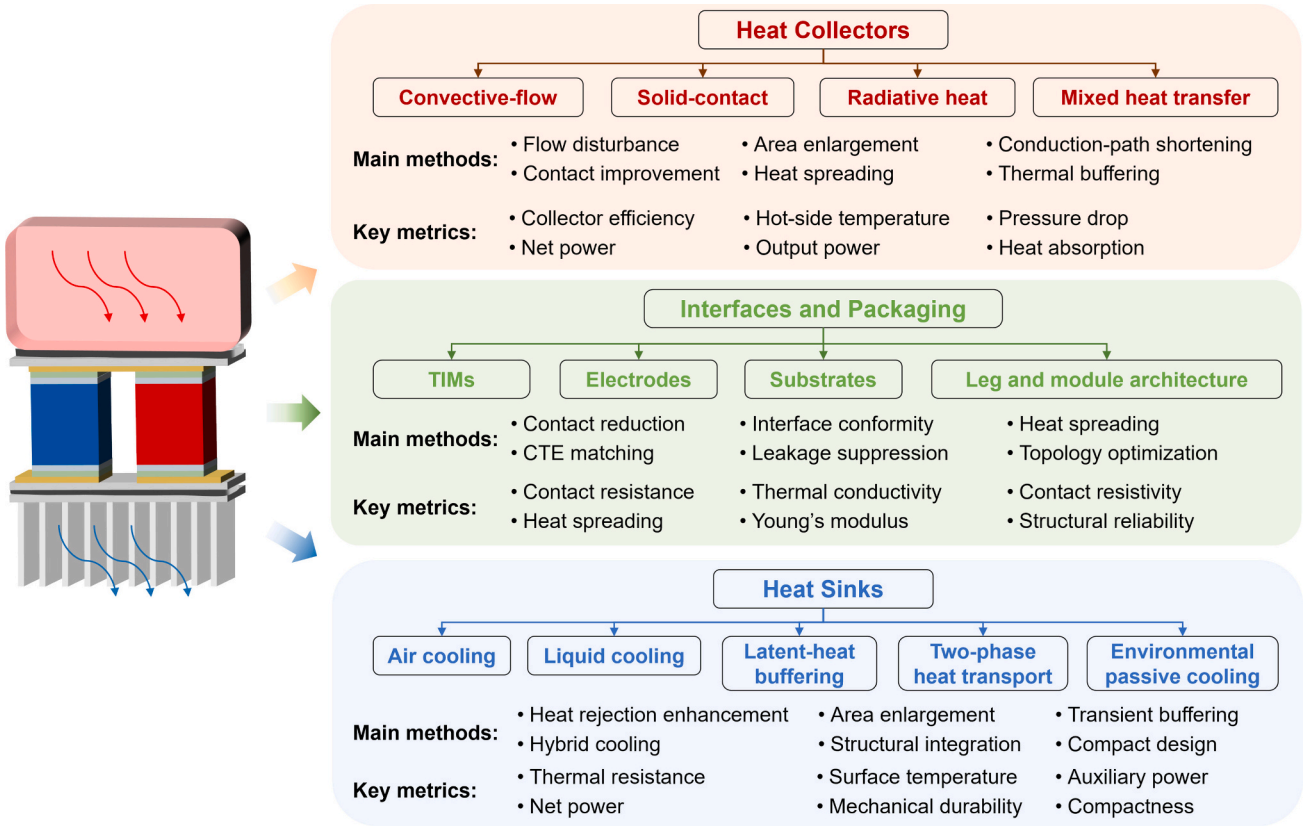

Fig. 15. The overall framework of TEG thermal management from the perspective of the thermal resistance chain.

6.2. *Net system gain under realistic operating conditions*

The second challenge is that local heat-transfer enhancement often fails to translate into net electrical gain. Intensified hot-side convection can increase exhaust backpressure; stronger forced cooling can consume more fan or pump power than the additional TEG output; microchannels, metal foams, vapor chambers, PCMs, and radiative/evaporative modules may introduce mass, volume, cost, maintenance, or environmental dependence. In flexible and wearable devices, additional cooling structures may also reduce breathability, conformability, and comfort.

Therefore, the relevant metric is not gross power enhancement but net system benefit. For active cooling systems, auxiliary fan, pump, and control power should be deducted. For automotive systems, pressure drop and engine backpressure should be included. For industrial systems, installation cost and disturbance to the host process should be considered. For wearable systems, power density should be reported together with thickness, areal mass, contact pressure, air permeability, and mechanical retention. For building-integrated systems, annual energy yield and thermal-comfort impact are more meaningful than short-term peak power.

Real operating conditions also differ strongly from steady-state laboratory boundaries. Exhaust temperature varies with driving cycles, solar irradiation changes with weather and time of day, combustion heat flux fluctuates with fuel feeding and fouling, geothermal sources experience local thermal drawdown, and wearable heat harvesting is affected by skin contact, motion, airflow, and humidity. Many advanced structures, including PCMs, heat pipes, vapor chambers, radiative coolers, and thermal buffers, should therefore be evaluated as dynamic thermal-capacity-matching devices. Their value may lie in extending generation duration, smoothing temperature fluctuations, suppressing thermal shock, or protecting the module, rather than only increasing instantaneous peak power.

Durability should also be treated as part of thermal-management performance. Interfaces can degrade through oxidation, interdiffusion, solder fatigue, and delamination. Heat collectors may suffer from fouling, ash deposition, corrosion, vibration, and thermal shock. Heat sinks may experience clogging, leakage, PCM degradation, hydrogel drying, or capillary failure. A design that gives high initial output but rapidly loses thermal contact or heat-transfer capacity cannot be considered a robust solution.

6.3. *Full-chain co-design and standardized reporting*

The third challenge is that thermal, electrical, and structural parameters are strongly coupled. Reducing external thermal resistance changes heat flux, module operating temperature, and optimal electrical load. Increasing leg length may preserve temperature difference but raise electrical resistance. A high-conductivity substrate can reduce through-plane resistance but cause lateral leakage in flexible systems. A powerful heat sink can improve ΔT but reduce net power if its auxiliary energy demand is excessive. These examples show that the optimal TEG system is not obtained by minimizing every resistance term independently.

Future TEG thermal management should therefore move toward full-chain co-design. Hot-side heat access, interface quality, module geometry, electrical load, cold-side heat rejection, parasitic leakage, mechanical stress, and reliability should be optimized together. For

variable operating conditions, adaptive strategies may also be useful, including load tracking, variable coolant flow, controllable thermal conductance, thermal bypass regulation, and predictive PCM charging/discharging management. The goal is to maximize net useful output over the target operating cycle, not simply to improve one local heat-transfer coefficient.

A more consistent reporting protocol is essential for this transition. At minimum, future studies should report far-field source and sink conditions, actual module hot-side and cold-side temperatures, heat flux or flow rate, contact pressure, interface material and thickness, pressure drop, auxiliary energy consumption, gross and net electrical output, temperature uniformity, test duration, transient profile, and durability conditions. For passive environmental systems, meteorological data, humidity, sky view factor, solar exposure, and cooling area should be included. For wearable systems, skin temperature, wearing pressure, airflow, bending state, thickness, areal mass, and on-body duration should be reported. Only with such data can interface designs, heat collectors, heat sinks, and integrated systems be compared fairly.

Overall, future TEG thermal management should follow a clear sequence: identify the dominant resistance, quantify the temperature-loss budget, evaluate local strategies under realistic penalties, test transient and long-term behavior, and optimize the full thermal-resistance chain for net system benefit. This transition from component-level enhancement to resistance-chain coordination is essential for converting thermal-management innovations into robust electrical output.

7. Conclusion

This review examined TEG thermal management from the perspective of parasitic thermal-resistance suppression and effective temperature-difference utilization. A generalized thermal-resistance network was first established to connect heat-source access, interface/package losses, thermoelectric heat transport, cold-side rejection, and parasitic heat leakage. Within this framework, recent advances in module interfaces and packaging, hot-side heat collectors, and cold-side heat sinks were reviewed. The central conclusion is that practical TEG performance is governed not only by thermoelectric material properties, but also by how much of the external source-to-sink temperature difference is preserved across the active legs.

The reviewed studies show that the dominant resistance changes with the heat-source boundary and system architecture. Flowing exhaust and industrial fluids are usually constrained by convection and pressure-drop penalties. Solid waste-heat systems are often limited by contact and spreading resistances. Solar TEGs require simultaneous control of optical access, hot-side loss, flux distribution, and environmental cooling. Geothermal systems are governed by heat-path length and thermal replenishment. Combustion-driven systems must stabilize mixed and contaminated thermal boundaries. Wearable systems are dominated by low-pressure contact, body-side conduction, lateral leakage, and comfort constraints. Therefore, there is no universal thermal-management structure that is optimal for all TEGs. The appropriate strategy must be selected after identifying the dominant resistance and its role in the full thermal-resistance chain.

A recurring lesson is that local thermal enhancement alone is insufficient. A stronger heat collector, TIM, heat sink, PCM buffer, or radiative cooler is valuable only if it increases net system output under realistic constraints. Future TEG research should therefore move from single-component optimization toward full-chain co-design. Dominant-resistance diagnosis, temperature-loss budgeting, net power evaluation, transient testing, durability assessment, and standardized reporting should become routine. By treating heat access, interface transfer, module design, heat rejection, leakage suppression, and auxiliary penalties as one coupled system, thermal-management research can more effectively translate local heat-transfer advances into reliable and scalable TEG performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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